

Polynomial–Logarithmic Transseries

Algebra, Division, Composition, and Asymptotic Series
Reversal

with Lambert’s W function as the guiding example

A self-contained computational monograph

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Abstract

Polynomial–logarithmic transseries are expansions whose successive power blocks carry polynomial, Laurent-series, or nested-logarithmic coefficients. The model example is the large-argument expansion of Lambert’s function,

$$W(x) = L_1 - L_2 + \frac{L_2}{L_1} + \frac{L_2^2/2 - L_2}{L_1^2} + \cdots, \quad L_1 = \log x, \quad L_2 = \log \log x.$$

Although such formulas look like ordinary power series with decorative logarithms, their correct algebra depends on an ordered hierarchy of scales: X^{-1} is smaller than every fixed negative power of $\log X$, while powers of $\log X$ remain ordered inside each X^{-n} block.

This article develops a practical and rigorous calculus for these objects. It defines the polynomial–logarithmic Laurent algebra and its iterated Laurent-field completion; proves closure and coefficient formulas for multiplication, reciprocal, and division; derives differentiation formulas that preserve the class; and gives two complementary constructions of composition. The first composes by substituting the generators X^{-1} and $\log X$; the second is a formally justified Taylor expansion for near-identity substitutions.

The central part treats series reversal, meaning compositional inversion. Fixed-point iteration, triangular coefficient recursion, Newton iteration, and a transserial Lagrange–Bürmann formula are developed and compared. For $F(X) = X + A(X)$ the inverse is shown formally to be

$$F^{(-1)}(Y) = Y + \sum_{n \geq 1} \frac{(-1)^n}{n!} \frac{d^{n-1}}{dY^{n-1}} A(Y)^n,$$

whenever the polynomial–logarithmic filtration makes the family summable. Taking $A(X) = \log X$ yields an all-orders derivation of the Lambert- W polynomials, their Stirling-number formula, and their integral recurrence. The discussion also covers complex branches, the lower real branch, scale-changing inversions such as $Y = cX^\rho(\log X)^\sigma$, nested logarithms, analytic error transfer, implementation strategies, and common failure modes.

The exposition is formal where formal algebra is the correct language and explicit about the additional hypotheses needed when a transseries is to be interpreted as an asymptotic expansion of an actual function.

Source and scope note

This monograph is an original synthesis based on the supplied archive, which contains TeX and PDF versions of Gerald Edgar’s foundational expositions on transseries and composition, later structural papers on logarithmic and surreal transseries, and several extensive tutorial drafts. Additional sources were checked online, especially Corless–Gonnet–Hare–Jeffrey–Knuth on Lambert W , the NIST Digital Library of Mathematical Functions, Salvy–Shackell on multiseriess inversion, van der Hoeven’s computational transseries theory, and the power–log transseries literature associated with Dulac series. Full references appear in the bibliography.

The phrase *polynomial–logarithmic transseries* is used here in a specific computational sense. The basic objects have the form

$$\sum_{n \geq n_0} X^{-n} P_n(\log X), \quad P_n \in \mathbb{K}[L],$$

possibly with fractional powers, Laurent-series coefficients in $1/\log X$, or finitely many iterated logarithmic levels. This is a useful subclass of logarithmic–exponential transseries, not a proposed replacement for the full theory. It is large enough for Lambert W , many inverse problems, quantile asymptotics, saddle-point calculations, and power–log normal-form calculations, while remaining concrete enough for exact coefficient algorithms.

How to read this article

Readers mainly interested in calculation can proceed as follows.

1. Read Chapters 1–2 for the scale and notation.
2. Use Chapters 3 and 4 for product, reciprocal, and quotient algorithms.
3. Use Chapters 7 and 8 for substitution.
4. Use Chapters 11–14 for compositional inverses.
5. Read Part V for the complete Lambert- W calculation.
6. Keep Appendix C open as a formula sheet.

The algebraic statements are given over a characteristic-zero coefficient field $\mathbb{K}$, usually $\mathbb{R}$ or $\mathbb{C}$. For an analytic interpretation, X tends to $+\infty$ unless a complex sector or another limit is explicitly stated.

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Part I

The polynomial–logarithmic scale

Chapter 1

Why powers and logarithms belong in one expansion

1.1 The prototype from Lambert's function

Lambert's W function is defined by

$$W(x)e^{W(x)} = x. \quad (1.1)$$

Taking a logarithm gives

$$W(x) + \log W(x) = \log x. \quad (1.2)$$

The dominant balance is $W(x) \sim \log x$, but substitution of this first approximation leaves the smaller error $\log \log x$. Thus the first two scales are forced:

$$W(x) = \log x - \log \log x + \text{smaller terms}.$$

If

$$L_1 = \log x, \quad L_2 = \log L_1,$$

then the complete expansion is organized in inverse powers of L_1 , with a polynomial in L_2 in each block:

$$W(x) = L_1 - L_2 + \sum_{n \geq 1} \frac{P_n(L_2)}{L_1^n}. \quad (1.3)$$

This block structure is the basic subject of the article.

1.2 The hierarchy inside a block expansion

Let $X \rightarrow +\infty$ and put

$$t = X^{-1}, \quad L = \log X.$$

Then

$$t \prec L^{-m} \prec 1 \prec L^m \prec t^{-1} \quad (m > 0).$$

More generally,

$$t^n L^k \succ t^m L^j \iff n < m \text{ or } (n = m \text{ and } k > j). \quad (1.4)$$

The power of t is compared first. Only when the t powers agree do we compare logarithmic powers. This is lexicographic asymptotic ordering.

For example,

$$\frac{L^{1000}}{X^3} \prec \frac{1}{X^2 L^{1000}},$$

because one additional inverse power of X defeats every fixed power of $\log X$. This elementary fact is the reason it is natural to store all powers of L at a fixed X^{-n} as one coefficient polynomial.

1.3 What “transseries” adds to an ordinary asymptotic series

An ordinary inverse-power expansion

$$a_0 + a_1 X^{-1} + a_2 X^{-2} + \dots$$

uses the scale $1, X^{-1}, X^{-2}, \dots$. A polynomial–logarithmic expansion uses the refined scale

$$\dots, X^{-n} L^2, X^{-n} L, X^{-n}, X^{-n} L^{-1}, \dots,$$

and may place another asymptotic series inside the coefficient of each X^{-n} . A full logarithmic–exponential transseries goes much further, allowing exponentials, nested exponentials, arbitrary well-based supports, and many levels of logarithms. The present class is a computationally transparent corner of that larger ordered field; see Edgar [3, 4] and van der Hoeven [11] for the ambient theory.

1.4 Relation to power–log and Dulac series

In local dynamics one often studies $x \rightarrow 0^+$ and writes

$$\ell(x) = -\frac{1}{\log x}.$$

Power–log transseries then use monomials $x^\alpha \ell^k$. Under $X = 1/x$,

$$x^\alpha \ell^k = X^{-\alpha} (\log X)^{-k}.$$

Thus the power–log notation and the present large- X notation encode the same two-scale geometry, with different choices of small generators. Dulac series and their extensions permit real exponent sets and logarithmic Laurent coefficients; see Mardešić–Resman–Rolin–Županović [7]. Our integer-lattice class is deliberately narrower at first, because this is where multiplication, division, composition, and reversion can be displayed with exact finite algorithms.

1.5 Formal equality, asymptotic equality, and analytic equality

Three notions must be kept separate.

1. A *formal identity* is an identity between coefficient arrays.
2. An *asymptotic expansion* states that finite truncations approximate a function in a specified limit.
3. An *analytic identity* is an equality of functions on a domain.

Formal manipulation does not by itself prove that a series converges. On the other hand, convergence is not required for formal coefficient extraction. Lambert W is unusually friendly: its standard double logarithmic series is not only asymptotic but is absolutely convergent in suitable real and complex domains; see Kalugin and Jeffrey [6], together with the classical treatment of Corless et al. [1] and the branch data collected in DLMF [2]. Most of the algebra developed below does not rely on this special fact.

Summary

Polynomial–logarithmic transseries are block expansions in which inverse powers of a large variable form the outer scale and powers or series in its logarithm form inner coefficients. The outer power is asymptotically decisive. The same formalism appears at 0 as power–log or Dulac-type series after a change of variables.

Chapter 2

The formal one-logarithm algebra

2.1 The polynomial-coefficient Laurent algebra

Let $\mathbb{K}$ be a field of characteristic zero. Introduce two formal symbols t and L , to be interpreted later as $t = X^{-1}$ and $L = \log X$.

Definition 2.1 (One-logarithm Laurent algebra). The basic polynomial-logarithmic algebra is

$$\mathcal{P}\mathcal{L}_{\text{poly}} = \mathbb{K}[L]((t)) = \left\{ \sum_{n \geq n_0} P_n(L) t^n : n_0 \in \mathbb{Z}, \right. \\ \left. P_n \in \mathbb{K}[L] \right\}. \quad (2.1)$$

Only finitely many negative powers of t are allowed, while infinitely many nonnegative powers may occur. Each coefficient P_n is an ordinary finite polynomial in L .

The representation is unique. It is useful to expand it further as

$$F = \sum_{n \geq n_0} \sum_{k=0}^{d_n} a_{n,k} t^n L^k.$$

The support is the set of pairs (n, k) for which $a_{n,k} \neq 0$. The asymptotic order is the lexicographic order in (1.4).

Definition 2.2 (Leading data). For nonzero $F \in \mathcal{P}\mathcal{L}_{\text{poly}}$, let n_0 be the smallest n with $P_n \neq 0$, and let $d = \deg P_{n_0}$. Then

$$\text{lm}(F) = t^{n_0} L^d, \quad \text{lc}(F) = [P_{n_0}]_{L^d}.$$

The pair $(n_0, -d)$ is the full lexicographic valuation of F .

2.2 Why the support is legitimate

At fixed n , only finitely many k occur. The outer exponents are bounded below. Consequently every nonempty support has a largest monomial in the asymptotic order, and each coefficient of a product receives only finitely many contributions. This is the elementary lattice version of the Hahn–Neumann support mechanism underlying general transseries.

Proposition 2.3 (Leading-term arithmetic). *For nonzero $F, G \in \mathcal{PL}_{\text{poly}}$,*

$$\text{lm}(FG) = \text{lm}(F) \text{lm}(G), \quad \text{lc}(FG) = \text{lc}(F) \text{lc}(G).$$

If $\text{lm}(F) \neq \text{lm}(G)$, then the larger of the two leading monomials is the leading monomial of $F + G$.

Proof. Write the leading terms as $at^n L^k$ and $bt^m L^j$. Their product $abt^{n+m} L^{k+j}$ is larger than every product involving a lower monomial of either factor. The assertion for sums is immediate unless the leading monomials agree, in which case cancellation may occur and one inspects the next term. $\square$

2.3 Block order and truncation

For coefficient algorithms we use the outer t -adic filtration.

Definition 2.4 (Block order). For $F = \sum_{n \geq n_0} P_n(L)t^n \neq 0$, define

$$\text{ord}_t F = n_0.$$

Write $F = \mathcal{O}_t(t^N)$ when $\text{ord}_t F \geq N$. This is a formal block statement, not the analytic estimate $|F| \leq C|t|^N$.

The truncation through block N is

$$[F]_{\leq N} = \sum_{n=n_0}^N P_n(L)t^n,$$

and

$$F \equiv G \pmod{t^{N+1}} \iff [F]_{\leq N} = [G]_{\leq N}.$$

All finite-order algorithms below take place exactly in the quotient algebra

$$\mathbb{K}[L]((t))/t^{N+1}\mathbb{K}[L][[t]],$$

with the finite principal part retained when needed.

2.4 Why a field completion is useful

The ring $\mathbb{K}[L]$ has only constant units. Thus $L + 1$ is not invertible in $\mathbb{K}[L]$, even though as an asymptotic object

$$\frac{1}{L+1} = L^{-1} - L^{-2} + L^{-3} - \dots$$

To make division systematic, enlarge the coefficient ring to the Laurent series field in L^{-1} :

$$\mathcal{C}_L = \mathbb{K}((L^{-1})), \quad \widehat{\mathcal{PL}} = \mathcal{C}_L((t)). \quad (2.2)$$

An element of $\mathcal{C}_L$ may contain finitely many positive powers of L and an infinite tail of negative powers. Therefore

$$\mathbb{K}[L] \subset \mathbb{K}[L, L^{-1}] \subset \mathbb{K}((L^{-1})).$$

The outer field $\widehat{\mathcal{PL}}$ is an iterated Laurent-series field and hence a field.

Remark 2.5. The order of iteration matters. In $\mathbb{K}((L^{-1}))((t))$, the generator t is smaller than every power of L^{-1} . This matches the asymptotic fact that $X^{-1} = o((\log X)^{-m})$ for every fixed m .

2.5 Fractional powers and general exponent lattices

If a problem contains $X^{-1/q}$, one may replace t by $s = X^{-1/q}$ and use $\mathbb{K}((L^{-1}))((s))$. More generally, for an additive subgroup $\Gamma \subset \mathbb{R}$ one may consider

$$\sum_{\gamma \in S} P_\gamma(L) X^{-\gamma},$$

where $S \subset \Gamma$ is well ordered and satisfies a finite-decomposition condition. Multiplication is then a Hahn-series convolution. Grid-based transseries place S inside a finitely generated additive semigroup, which is especially convenient for computation [3, 4]. The integer-lattice notation will be used in the main derivations, since the same formulas then extend verbatim after replacing integer indices by admissible exponents.

2.6 Nested logarithmic levels

Let

$$L_0 = X, \quad L_{j+1} = \log L_j, \quad \varepsilon_j = L_j^{-1}.$$

For fixed depth d , a natural iterated field is

$$\mathbb{K}((\varepsilon_d)) \cdots ((\varepsilon_2))((\varepsilon_1)). \quad (2.3)$$

The rightmost variable ε_1 is the outermost and smallest scale. In particular,

$$\varepsilon_1 \prec \varepsilon_2^m \prec \varepsilon_3^n \prec 1 \quad (m, n > 0).$$

The Lambert expansion (1.3) belongs to the subring

$$\mathbb{K}[L_2]((L_1^{-1}))$$

with base variable $X = L_1$ and logarithm $L = L_2$.

2.7 Asymptotic interpretation and uniqueness

For actual functions, a convenient one-layer definition is the following.

Definition 2.6 (Block asymptotic expansion). We write

$$f(X) \sim_{\mathcal{P}\mathcal{L}} \sum_{n \geq n_0} P_n(\log X) X^{-n} \quad (2.4)$$

if for every integer $N \geq n_0$,

$$f(X) - \sum_{n=n_0}^N P_n(\log X) X^{-n} = o(X^{-N}) \quad (X \rightarrow +\infty).$$

Proposition 2.7 (Uniqueness of polynomial blocks). *An expansion in the sense of (2.4) has unique coefficient polynomials P_n .*

Proof. Suppose two expansions first differ at block n . Their difference divided by X^{-n} is a nonzero polynomial $Q(\log X) + o(1)$. A nonzero polynomial in $\log X$ cannot tend to zero as $X \rightarrow \infty$, a contradiction. $\square$

For coefficients in $\mathbb{K}((L^{-1}))$, one uses nested truncations: first truncate in t , then within each retained coefficient truncate in L^{-1} . This is the analytic counterpart of the iterated field (2.2).

Summary

The elementary polynomial–logarithmic ring is $\mathbb{K}[L]((t))$ with $t = X^{-1}$ and $L = \log X$. It is an ordered integral domain and a complete t -adic algebra, but not a field. The iterated Laurent field $\mathbb{K}((L^{-1}))((t))$ supplies the missing inverses while preserving the rule that t is smaller than every power of L^{-1} .

Part II

Arithmetic and differential calculus

Chapter 3

Multiplication: convolution by logarithmic blocks

3.1 The block Cauchy product

Let

$$F = \sum_{n \geq n_0} A_n(L)t^n, \quad G = \sum_{m \geq m_0} B_m(L)t^m. \quad (3.1)$$

Their product is obtained by the ordinary Cauchy rule in the outer variable:

$$FG = \sum_{N \geq n_0 + m_0} C_N(L)t^N, \quad C_N(L) = \sum_{n+m=N} A_n(L)B_m(L). \quad (3.2)$$

For a fixed N , the restrictions $n \geq n_0$ and $m \geq m_0$ leave only finitely many pairs. Hence (3.2) is an exact algebraic formula, not a rearrangement of an analytically convergent double series.

Theorem 3.1 (Closure and finite determination). *The algebra $\mathcal{P}\mathcal{L}_{\text{poly}} = \mathbb{K}[L]((t))$ is closed under multiplication. Moreover, the coefficient of t^N in FG depends only on the blocks*

$$A_{n_0}, \dots, A_{N-m_0}, \quad B_{m_0}, \dots, B_{N-n_0}.$$

Consequently, if $F \equiv \tilde{F} \pmod{t^p}$ and $G \equiv \tilde{G} \pmod{t^q}$, then

$$FG - \tilde{F}\tilde{G} = \mathcal{O}_t\left(t^{\min(p+m_0, q+n_0)}\right). \quad (3.3)$$

Proof. Formula (3.2) is a finite sum of products of polynomials and therefore gives $C_N \in \mathbb{K}[L]$. The support remains bounded below by $n_0 + m_0$. For the error rule, write $F = \tilde{F} + E$ and $G = \tilde{G} + H$, with $\text{ord}_t E \geq p$ and $\text{ord}_t H \geq q$. Then

$$FG - \tilde{F}\tilde{G} = E\tilde{G} + \tilde{F}H + EH,$$

and the three outer orders are at least $p + m_0$, $q + n_0$, and $p + q$. The first two bounds give (3.3). $\square$

3.2 The coefficient-level convolution

Write

$$A_n(L) = \sum_{i=0}^{a_n} a_{n,i} L^i, \quad B_m(L) = \sum_{j=0}^{b_m} b_{m,j} L^j.$$

Then

$$[t^N L^k](FG) = \sum_{n+m=N} \sum_{i+j=k} a_{n,i} b_{m,j}. \quad (3.4)$$

Thus multiplication is a two-dimensional convolution, but it is normally more efficient to perform it in two levels: first convolve the outer blocks, then multiply the coefficient polynomials. This respects the asymptotic hierarchy and allows a separate cutoff in L^{-1} if the coefficient field is $\mathbb{K}((L^{-1}))$.

Proposition 3.2 (Degree and support bounds). *For the product coefficient in (3.2),*

$$\deg_{\mathbb{L}} C_N \leq \max_{n+m=N} (\deg_{\mathbb{L}} A_n + \deg_{\mathbb{L}} B_m).$$

If all A_n and B_m have nonnegative coefficients over an ordered field, then equality holds whenever the maximizing blocks are nonzero.

Proof. Each summand has degree at most the displayed sum of degrees. Cancellation can lower the degree in general. With nonnegative coefficients there is no cancellation at the largest power of L . $\square$

3.3 A complete multiplication example

Take

$$\begin{aligned} F &= 1 + (L+1)t + (L^2 - L)t^2 + (2L+3)t^3 + \mathcal{O}_t(t^4), \\ G &= 1 - Lt + (2L-1)t^2 + (L^2+1)t^3 + \mathcal{O}_t(t^4). \end{aligned}$$

The blocks are computed one at a time:

$$\begin{aligned} C_0 &= 1, \\ C_1 &= (L+1) - L = 1, \\ C_2 &= (L^2 - L) + (L+1)(-L) + (2L-1) = L-1, \\ C_3 &= (2L+3) + (L^2 - L)(-L) + (L+1)(2L-1) + (L^2+1) \\ &= -L^3 + 4L^2 + 2L + 3. \end{aligned}$$

Therefore

$$FG = 1 + t + (L-1)t^2 + (-L^3 + 4L^2 + 2L + 3)t^3 + \mathcal{O}_t(t^4). \quad (3.5)$$

The calculation illustrates why a block representation is preferable to a flat list of monomials: the outer convolution is transparent, and all logarithmic algebra is delegated to ordinary polynomial arithmetic.

3.4 Truncated multiplication algorithm

Suppose both factors begin at block zero and are wanted through t^N .

Input: Blocks $A_0, \dots, A_N$ and $B_0, \dots, B_N$ in a coefficient ring

Output: Blocks $C_0, \dots, C_N$ of $FG \bmod t^{N+1}$

```

for  $r \leftarrow 0$  to  $N$  do
   $C_r \leftarrow 0$ ;
  for  $i \leftarrow 0$  to  $r$  do
     $C_r \leftarrow C_r + A_i B_{r-i}$ ;
  end
end
return  $(C_0, \dots, C_N)$ ;

```

Algorithm 1: Truncated polynomial–logarithmic product

With schoolbook polynomial multiplication, the arithmetic cost is governed by both the number of blocks and their logarithmic degrees. If the typical coefficient degree is d , the direct cost is $O(N^2 d^2)$ scalar operations. Fast convolution and fast polynomial multiplication can reduce this, but for asymptotic calculations the degrees and truncation orders are often modest, and sparse storage is more important than asymptotic complexity.

3.5 Multiplication in the completed coefficient field

Let

$$A_n(L) = \sum_{j \geq j_n} a_{n,j} L^{-j}, \quad B_m(L) = \sum_{k \geq k_m} b_{m,k} L^{-k}.$$

At a fixed outer block N , the product coefficient C_N is a finite sum of Laurent-series products. At a fixed inner power L^{-r} , each such product has only finitely many decompositions $j + k = r$. Thus the iterated order “first t , then L^{-1} ” makes multiplication coefficientwise finite at both levels.

Warning 3.3. A rectangular cutoff in the exponent plane need not respect asymptotic order. For example, discarding all terms with $|k| > 10$ may remove $t^2 L^{100}$ while retaining the much smaller $t^3 L^{-100}$. The safe strategy is a block cutoff in t , followed by a separately declared cutoff inside each coefficient.

3.6 Products of asymptotic expansions

Formal multiplication mirrors a standard analytic rule.

Proposition 3.4 (Analytic product rule). *Assume*

$$f(X) = \sum_{n=n_0}^N A_n(L) X^{-n} + o(X^{-N}), \quad g(X) = \sum_{m=m_0}^M B_m(L) X^{-m} + o(X^{-M}),$$

where the displayed finite sums have at most polynomial growth in L . Then for every R below the available precision,

$$f(X)g(X) = \sum_{n+m \leq R} A_n(L) B_m(L) X^{-n-m} + o(X^{-R}).$$

Proof. Multiply the two finite expansions and collect blocks. A discarded term X^{-p} times a polynomial in L is $o(X^{-p+\delta})$ for every $\delta > 0$. Taking the input expansions sufficiently far beyond the desired integer block R absorbs all logarithmic factors. The stated block hypothesis, which uses exact integer powers, then yields the result. $\square$

Summary

Multiplication is ordinary Cauchy convolution in $t = X^{-1}$ with polynomial or Laurent-series arithmetic in $L = \log X$. Every retained block receives only finitely many contributions. Truncation should always follow the iterated asymptotic hierarchy rather than an arbitrary rectangular exponent cutoff.

Chapter 4

Reciprocals and division

4.1 Which elements are units?

A nonzero element of the field $\widehat{\mathcal{PL}} = \mathbb{K}((L^{-1}))((t))$ is invertible. In the smaller ring $\mathcal{PL}_{\text{poly}} = \mathbb{K}[L]((t))$, invertibility is more restrictive.

Proposition 4.1 (Units of the polynomial coefficient ring). *Let*

$$F = t^\nu (A_0(L) + A_1(L)t + A_2(L)t^2 + \cdots), \quad A_0 \neq 0.$$

Then F is a unit of $\mathbb{K}[L]((t))$ if and only if A_0 is a nonzero constant. It is always a unit of $\mathbb{K}((L^{-1}))((t))$.

Proof. The factor t^ν is invertible. In a formal power-series ring $R[[t]]$, a series is a unit precisely when its constant coefficient is a unit of R . The only units of $\mathbb{K}[L]$ are nonzero constants, whereas every nonzero element of the field $\mathbb{K}((L^{-1}))$ is a unit. $\square$

Thus division may enlarge the coefficient class. For example,

$$\frac{1}{L+1} = L^{-1} - L^{-2} + L^{-3} - \cdots$$

is perfectly meaningful in $\mathbb{K}((L^{-1}))$, but not in $\mathbb{K}[L]$.

4.2 Normalization and the geometric expansion

Every nonzero $B \in \widehat{\mathcal{PL}}$ can be written uniquely as

$$B = b t^\nu L^\mu (1 + S), \quad b \in \mathbb{K}^\times, \quad S \prec 1. \quad (4.1)$$

Then

$$B^{-1} = b^{-1} t^{-\nu} L^{-\mu} \sum_{r \geq 0} (-S)^r. \quad (4.2)$$

The support conditions guarantee that the geometric family is summable: for each retained monomial, only finitely many powers of S can contribute. Formula (4.2) is the universal conceptual construction. For finite computation, a coefficient recurrence is usually faster.

4.3 The triangular quotient recurrence

First assume

$$A = \sum_{n \geq 0} a_n t^n, \quad B = \sum_{n \geq 0} b_n t^n, \quad b_0 \neq 0, \quad (4.3)$$

with coefficients in a field such as $\mathbb{K}((L^{-1}))$. Write $Q = A/B = \sum_{n \geq 0} q_n t^n$. The identity $BQ = A$ gives

$$q_0 = \frac{a_0}{b_0}, \quad q_N = \frac{1}{b_0} \left(a_N - \sum_{j=1}^N b_j q_{N-j} \right), \quad N \geq 1. \quad (4.4)$$

This is triangular: once $q_0, \dots, q_{N-1}$ are known, q_N follows by one coefficient-field division.

For a reciprocal $R = B^{-1} = \sum r_n t^n$, set $a_0 = 1$ and $a_N = 0$ for $N \geq 1$:

$$r_0 = b_0^{-1}, \quad r_N = -b_0^{-1} \sum_{j=1}^N b_j r_{N-j}. \quad (4.5)$$

Theorem 4.2 (Correctness of truncated division). *If $q_0, \dots, q_N$ are defined by (4.4), then*

$$B \sum_{n=0}^N q_n t^n \equiv A \pmod{t^{N+1}}.$$

They are the unique blocks with this property.

Proof. The coefficient of t^N in BQ is $b_0 q_N + \sum_{j=1}^N b_j q_{N-j}$, which equals a_N by construction. Induction gives all blocks through N . If two truncated quotients worked, their difference would have a first nonzero block ct^r ; multiplication by B would have first block $b_0 ct^r \neq 0$, a contradiction. $\square$

4.4 A reciprocal example

Let

$$B = 1 - Lt + (2L - 1)t^2 + (L^2 + 1)t^3 + \mathcal{O}_t(t^4).$$

Applying (4.5) gives

$$\begin{aligned} r_0 &= 1, \\ r_1 &= L, \\ r_2 &= L^2 - 2L + 1, \\ r_3 &= L^3 - 5L^2 + 2L - 1, \\ r_4 &= L^4 - 8L^3 + 7L^2 - 6L + 1, \end{aligned}$$

where r_4 is determined by taking the omitted t^4 block of B to be zero. Hence

$$\begin{aligned} B^{-1} &= 1 + Lt + (L^2 - 2L + 1)t^2 + (L^3 - 5L^2 + 2L - 1)t^3 \\ &\quad + (L^4 - 8L^3 + 7L^2 - 6L + 1)t^4 + \mathcal{O}_t(t^5). \end{aligned} \quad (4.6)$$

Multiplication by B checks the result block by block.

4.5 A quotient example

With

$$\begin{aligned} A &= 1 + (L + 1)t + (L^2 - L)t^2 + (2L + 3)t^3 + \mathcal{O}_t(t^4), \\ B &= 1 - Lt + (2L - 1)t^2 + (L^2 + 1)t^3 + \mathcal{O}_t(t^4), \end{aligned}$$

recurrence (4.4) yields

$$\begin{aligned} \frac{A}{B} &= 1 + (2L + 1)t + (3L^2 - 2L + 1)t^2 \\ &\quad + (3L^3 - 7L^2 + 3L + 3)t^3 + \mathcal{O}_t(t^4). \end{aligned} \quad (4.7)$$

A reliable implementation should verify the quotient by multiplying the right-hand side by the divisor to the requested order. This residual check is inexpensive and catches most cutoff or normalization mistakes.

4.6 Leading monomials other than one

For general A and B , first factor the leading monomial of B :

$$B = bt^\nu L^\mu (1 + S).$$

Then

$$\frac{A}{B} = b^{-1} t^{-\nu} L^{-\mu} A (1 + S)^{-1}.$$

If the leading coefficient is a nonmonomial Laurent series $b_0(L)$, invert it in $\mathbb{K}((L^{-1}))$ before applying the outer recurrence. For example,

$$\begin{aligned} \frac{1}{L^2 + 3L + 2} &= L^{-2} \frac{1}{1 + 3L^{-1} + 2L^{-2}} \\ &= L^{-2} - 3L^{-3} + 7L^{-4} - 15L^{-5} + 31L^{-6} + \dots \end{aligned}$$

There are therefore two nested divisions: an inner Laurent-series inversion for the leading coefficient and an outer t -series inversion.

4.7 Newton inversion

For high order, Newton iteration replaces the linear recurrence by a precision-doubling scheme. If R approximates B^{-1} , set

$$R_{\text{new}} = R(2 - BR). \quad (4.8)$$

If $BR = 1 - E$, then

$$BR_{\text{new}} = (1 - E)(1 + E) = 1 - E^2.$$

Thus an error of outer order m becomes one of order $2m$. Starting with $R_0 = b_0^{-1}$, one truncates after each step to orders 1, 2, 4, 8, ... until the target precision is reached.

Algorithmic principle 4.3 (Guard precision). When a quotient will later be multiplied by a factor of negative outer order, compute it beyond the final requested block. Truncation is safe only after propagating valuations through the entire expression tree. A robust system uses explicit precision objects rather than a single global cutoff.

4.8 Division of asymptotic expansions

Suppose $g(X)$ has a nonzero leading block and is bounded away from zero after factoring its leading monomial. If formal division gives $Q_N(X)$ through block N and

$$f(X) - g(X)Q_N(X) = o(X^{-N}),$$

then

$$\frac{f(X)}{g(X)} - Q_N(X) = \frac{o(X^{-N})}{g(X)}.$$

The leading scale of g converts this residual into the corresponding quotient error. This simple residual argument is often more reliable than trying to manipulate analytic remainder symbols at every intermediate step.

Summary

Normalize a divisor by its leading monomial and invert the remaining unit. At finite order, quotient blocks satisfy a triangular recurrence. Newton's formula $R \mapsto R(2 - BR)$ doubles precision and is preferable at high order. Division may force polynomial logarithmic coefficients to expand into Laurent series in $1/\log X$.

Chapter 5

Powers, logarithms, exponentials, and closure boundaries

5.1 General powers of a normalized element

Let

$$F = c t^\nu L^\mu (1 + S), \quad S \prec 1. \quad (5.1)$$

For $\alpha \in \mathbb{K}$ one formally defines

$$F^\alpha = c^\alpha t^{\alpha\nu} L^{\alpha\mu} \sum_{r \geq 0} \binom{\alpha}{r} S^r. \quad (5.2)$$

Over $\mathbb{R}$, a real value for c^α requires the usual sign conditions. Over $\mathbb{C}$, a branch of $\text{Log } c$ must be selected. Rational α may force a fractional outer exponent lattice; irrational α requires a more general ordered exponent group.

Proposition 5.1 (Finite determination of a binomial power). *If $S = \mathcal{O}_t(t^q)$ with $q > 0$, then the coefficient of t^N in $(1 + S)^\alpha$ receives contributions only from powers $r \leq \lfloor N/q \rfloor$. Hence (5.2) is coefficientwise finite.*

Proof. The outer order of S^r is at least rq . □

5.2 Logarithms

For (5.1),

$$\log F = \log c + \nu \log t + \mu \log L + \log(1 + S). \quad (5.3)$$

Since $\log t = -L$, this becomes

$$\log F = \log c - \nu L + \mu \log L + \sum_{r \geq 1} \frac{(-1)^{r+1}}{r} S^r. \quad (5.4)$$

Two closure facts are visible.

1. If $\mu = 0$ and S is t -adically small, the logarithm stays in the one-logarithm class, apart from the allowed linear term in L .
2. If $\mu \neq 0$, the new generator $\log L = \log \log X$ appears. The one-logarithm class is therefore not closed under logarithm, but the finite nested-logarithm tower is.

Example 5.2. For $F = X^3 L^{-2}(1 + (L + 1)/X)$,

$$\begin{aligned}\log F &= 3L - 2 \log L + \log(1 + (L + 1)t) \\ &= 3L - 2 \log L + (L + 1)t - \frac{1}{2}(L + 1)^2 t^2 + \frac{1}{3}(L + 1)^3 t^3 + \cdots.\end{aligned}$$

The nested logarithm is forced by the factor L^{-2} , not by the small correction.

5.3 Exponentials

If $S \prec 1$, then

$$\exp S = \sum_{r \geq 0} \frac{S^r}{r!} \quad (5.5)$$

is summable and begins with 1. If $F = F_{\text{large}} + c + S$ is decomposed into a purely large part, constant, and small part, then

$$\exp F = e^c \exp(F_{\text{large}}) \exp S.$$

The factor $\exp(F_{\text{large}})$ is generally a new transmonomial. For example,

$$\exp(L) = X = t^{-1}$$

stays inside the class, but $\exp(L^2) = \exp((\log X)^2)$ does not belong to $\mathbb{K}((L^{-1}))((t))$. It belongs naturally to a larger logarithmic-exponential transseries field.

Pitfall

The statement “ S is small” must refer to the asymptotic ordering, not to small numerical coefficients. The formal exponential of L^{-1} is a unit, whereas the exponential of L^2 creates a new dominant scale. Treating both with the same flat power-series routine destroys the monomial hierarchy.

5.4 Analytic functions of a unit

Let $\Phi(z) = \sum_{r \geq 0} c_r (z - z_0)^r$ be analytic or merely formal at z_0 , and let $F = z_0 + S$ with $S \prec 1$. Then

$$\Phi(F) = \sum_{r \geq 0} c_r S^r \quad (5.6)$$

is a valid transseries substitution whenever the powers are summable. This single principle supplies square roots, trigonometric functions of small arguments, rational powers, and local inverse functions.

For example,

$$\begin{aligned}(1 + S)^{1/2} &= 1 + \frac{1}{2}S - \frac{1}{8}S^2 + \frac{1}{16}S^3 - \cdots, \\ \frac{1}{1 + S} &= 1 - S + S^2 - S^3 + \cdots, \\ \log(1 + S) &= S - \frac{1}{2}S^2 + \frac{1}{3}S^3 - \cdots.\end{aligned}$$

All three use the same support argument.

5.5 A closure table

Operation	Typical hypothesis	Resulting class
Addition, multiplication	none beyond support conditions	$\mathbb{K}[L]((t))$
Reciprocal	leading coefficient constant	$\mathbb{K}[L]((t))$
General reciprocal	nonzero leading coefficient	$\mathbb{K}((L^{-1}))((t))$
Derivative	any polynomial–log series	same one-logarithm class
$\Phi(c + S)$	$S \prec 1$, Φ local formal series	same class if coefficients permit
Logarithm	normalized $ct^\nu L^\mu(1 + S)$	adds $\log L$ when $\mu \neq 0$
Exponential	small argument	a unit in the same class
Exponential	nontrivial large part	may require new exponential monomials
Composition	large admissible inner argument	may require fractional powers or deeper logs

Summary

Normalize before applying nonlinear operations. Binomial, logarithmic, and exponential series are safe on asymptotically small units. Logarithms of logarithmic powers create deeper logarithms, and exponentials of genuinely large polynomial–logarithmic expressions usually leave the class. These are structural closure boundaries, not implementation defects.

Chapter 6

Differentiation and antiderivatives

6.1 The basic derivation

Recall $t = X^{-1}$ and $L = \log X$. Therefore

$$\frac{dt}{dX} = -t^2, \quad \frac{dL}{dX} = t.$$

On a coefficient block this gives the fundamental formula

$$\mathcal{D}(P(L)t^n) := \frac{d}{dX}(P(L)X^{-n}) = t^{n+1}(P'(L) - nP(L)). \quad (6.1)$$

Equivalently,

$$\mathcal{D} = -t^2 \frac{\partial}{\partial t} + t \frac{\partial}{\partial L}. \quad (6.2)$$

Proposition 6.1 (Closure under differentiation). *The derivation $\mathcal{D}$ maps $\mathbb{K}[L]((t))$ into itself and raises outer order by at least one unless cancellation occurs in a leading coefficient. It extends termwise to $\mathbb{K}((L^{-1}))((t))$.*

Proof. Formula (6.1) maps a polynomial to a polynomial and sends block n to block $n + 1$. For Laurent-series coefficients, differentiation with respect to L also preserves $\mathbb{K}((L^{-1}))$. $\square$

6.2 The Euler derivation

It is often cleaner to use

$$\Theta = X \frac{d}{dX} = t^{-1} \mathcal{D}. \quad (6.3)$$

Then

$$\Theta(P(L)t^n) = t^n(P' - nP). \quad (6.4)$$

Thus each outer block is invariant under Θ , and its coefficient is acted on by the first-order operator $\partial_L - n$. This diagonal block structure is useful in differential equations.

6.3 Repeated derivatives

Iterating (6.1) gives a compact formula.

Theorem 6.2 (Repeated derivative formula). *For $r \geq 0$,*

$$\mathcal{D}^r(P(L)t^n) = t^{n+r} \prod_{j=0}^{r-1} (\partial_L - (n+j))P(L), \quad (6.5)$$

where the empty product is the identity.

Proof. The case $r = 0$ is trivial. If the formula holds for r , apply (6.1) to its coefficient at outer exponent $n + r$. This appends the commuting factor $\partial_L - (n + r)$. $\square$

In particular,

$$\begin{aligned} \mathcal{D}^r(L^m) &= t^r \prod_{j=0}^{r-1} (\partial_L - j)L^m, \\ \mathcal{D}^r(t^n) &= (-1)^r (n)^{\bar{r}} t^{n+r}, \end{aligned}$$

where $(n)^{\bar{r}} = n(n+1) \cdots (n+r-1)$ is the rising factorial. The first identity is the source of the Stirling-number polynomials in the Lambert expansion.

6.4 Product and chain rules

Because (6.2) is an ordinary derivation on the formal generators, it satisfies

$$\mathcal{D}(FG) = (\mathcal{D}F)G + F(\mathcal{D}G).$$

Whenever composition is admissible,

$$\mathcal{D}(F \circ G) = (\mathcal{D}F \circ G)\mathcal{D}G. \quad (6.6)$$

The chain rule can be verified first for the generators t and L , then for finite algebraic expressions, and finally extended by coefficientwise summability. In a full transseries field it is one of the central compatibility properties of composition [4, 11].

6.5 Solving for an antiderivative block

To integrate $Q(L)t^m$ with $m \neq 1$, seek an antiderivative of the form $P(L)t^{m-1}$. Equation (6.1) requires

$$P' - (m-1)P = Q. \quad (6.7)$$

If Q is a polynomial and $m \neq 1$, this equation has a unique polynomial solution. Indeed,

$$P = - \sum_{j=0}^{\deg Q} \frac{Q^{(j)}}{(m-1)^{j+1}}. \quad (6.8)$$

Differentiating the finite sum verifies (6.7). Therefore

$$\int Q(L)X^{-m} dX = X^{1-m} \left[- \sum_{j=0}^{\deg Q} \frac{Q^{(j)}(L)}{(m-1)^{j+1}} \right] + C, \quad m \neq 1. \quad (6.9)$$

When $m = 1$, equation (6.7) becomes $P' = Q$, so

$$\int Q(L)X^{-1} dX = \int Q(L) dL,$$

again polynomial. Hence the polynomial-logarithmic ring is closed under termwise antiderivatives, up to an arbitrary constant, in this one-level integer-exponent setting.

Example 6.3. Since

$$\int \frac{L^2}{X^3} dX = X^{-2}P(L),$$

we solve $P' - 2P = L^2$ and obtain

$$P = -\frac{1}{2}L^2 - \frac{1}{2}L - \frac{1}{4}.$$

Thus

$$\int \frac{(\log X)^2}{X^3} dX = -\frac{2L^2 + 2L + 1}{4X^2} + C.$$

6.6 Differential equations by block matching

Consider

$$Y' + Y = \frac{L}{X}. \quad (6.10)$$

Seek $Y = \sum_{n \geq 1} P_n(L)t^n$. From (6.1),

$$Y' + Y = P_1 t + \sum_{n \geq 2} (P_n + P'_{n-1} - (n-1)P_{n-1})t^n.$$

Hence $P_1 = L$ and

$$P_n = (n-1)P_{n-1} - P'_{n-1}, \quad n \geq 2.$$

The first blocks are

$$Y = Lt + (L-1)t^2 + (2L-3)t^3 + (6L-11)t^4 + \dots$$

This example shows how differentiation's shift by one outer block makes many linear differential equations triangular.

6.7 Why differentiation controls composition and inversion

If $A = \mathcal{O}_t(1)$ contains no X term, then

$$\mathcal{D}A = \mathcal{O}_t(t). \quad (6.11)$$

Consequently, changing an argument by an error $E = \mathcal{O}_t(t^r)$ changes A by roughly $A'E = \mathcal{O}_t(t^{r+1})$. This one-block gain is the mechanism behind the contraction used for compositional inversion. Repeated derivatives also show that the n th Lagrange–Bürmann term contributes no earlier than block $n-1$.

Summary

The derivative of $P(L)X^{-n}$ is $X^{-n-1}(P' - nP)$. Repeated derivatives factor into commuting operators $\partial_L - (n+j)$. Differentiation raises outer order, a fact that makes near-identity Taylor composition and Lagrange reversion coefficientwise finite. Polynomial-logarithmic blocks also admit explicit polynomial antiderivatives.

Part III

Composition

Chapter 7

Composition by substitution of the generators

7.1 Why composition is the delicate operation

Addition and multiplication combine supports already present. Composition changes the scale itself: replacing X by $G(X)$ transforms both X^{-1} and $\log X$, and the transformed generators may introduce fractional powers, new logarithmic levels, or entirely new exponential monomials. In the full transseries field, composition is defined for a large positive inner transseries and is constrained by summability; Edgar’s composition paper gives a detailed treatment [4]. In the polynomial–logarithmic setting, the most important and cleanest case is a near-identity argument.

7.2 Near-identity arguments

Let

$$G(X) = X + \Delta(X) = X(1 + U(X)), \quad U = t\Delta. \quad (7.1)$$

Assume

$$U \in t\mathcal{C}_L[[t]], \quad (7.2)$$

so $U \prec 1$. The two formal generators transform exactly as

$$t = \frac{1}{X} \mapsto \frac{1}{G} = t(1 + U)^{-1}, \quad L = \log X \mapsto \log G = L + \log(1 + U). \quad (7.3)$$

Both right-hand sides belong to $\widehat{\mathcal{P}\mathcal{L}}$ and are computable by geometric and logarithmic series.

Definition 7.1 (Near-identity composition). For

$$F(X) = \sum_{n \geq n_0} P_n(L) t^n$$

and G satisfying (7.1)–(7.2), define

$$(F \circ G)(X) = \sum_{n \geq n_0} t^n (1 + U)^{-n} P_n(L + \log(1 + U)). \quad (7.4)$$

Theorem 7.2 (Well-definedness at finite order). *For every outer cutoff N , the truncation of (7.4) through t^N depends on only finitely many input blocks and finitely many terms of each binomial, logarithmic, and polynomial Taylor expansion. Therefore near-identity composition is a well-defined operation on $\widehat{\mathcal{PL}}$.*

Proof. Because $U = \mathcal{O}_t(t)$, the coefficient of t^r in $(1 + U)^{-n}$ or $\log(1 + U)$ depends on finitely many powers of U . A coefficient polynomial P_n has finite degree, so its substitution at $L + \log(1 + U)$ is a finite Taylor polynomial in the logarithmic increment. Finally, only blocks $n \leq N$ can affect the output through t^N . $\square$

7.3 A direct calculation

Take

$$F(X) = \log X + \frac{1}{X} = L + t, \quad G(X) = X + \log X = X + L.$$

Here $U = Lt$. The transformed generators are

$$\begin{aligned} \log G &= L + \log(1 + Lt) \\ &= L + Lt - \frac{1}{2}L^2t^2 + \frac{1}{3}L^3t^3 - \frac{1}{4}L^4t^4 + \mathcal{O}_t(t^5), \\ \frac{1}{G} &= t(1 + Lt)^{-1} \\ &= t - Lt^2 + L^2t^3 - L^3t^4 + \mathcal{O}_t(t^5). \end{aligned}$$

Therefore

$$F \circ G = L + (L + 1)t - \left(L + \frac{1}{2}L^2\right)t^2 + \left(L^2 + \frac{1}{3}L^3\right)t^3 - \left(L^3 + \frac{1}{4}L^4\right)t^4 + \mathcal{O}_t(t^5). \quad (7.5)$$

One may check the same result by Taylor composition in the next chapter.

7.4 Composition as an algebra homomorphism

For fixed admissible G , substitution respects the ring operations:

$$(F_1 + F_2) \circ G = (F_1 \circ G) + (F_2 \circ G), \quad (7.6)$$

$$(F_1 F_2) \circ G = (F_1 \circ G)(F_2 \circ G). \quad (7.7)$$

It also respects reciprocal whenever the factor is nonzero:

$$(F^{-1}) \circ G = (F \circ G)^{-1}.$$

These statements follow because (7.3) defines a homomorphism from the algebra generated by t, L and summable expansions into itself.

7.5 Associativity

Suppose G and H are near the identity and all indicated substitutions remain in the same admissible class. Then

$$(F \circ G) \circ H = F \circ (G \circ H). \quad (7.8)$$

Indeed, both sides replace X by $G(H(X))$. On the generators,

$$\frac{1}{X} \mapsto \frac{1}{G(H(X))}, \quad \log X \mapsto \log(G(H(X))),$$

and homomorphic extension gives equality. The qualification about admissibility matters in larger Hahn fields: a syntactically meaningful nested expression need not define a summable support without hypotheses.

7.6 Practical generator-substitution algorithm

To compute $F \circ G$ through block N :

1. Form $U = t(G - X)$ through block $N + 1$.
2. Compute $R = (1 + U)^{-1}$ and $V = \log(1 + U)$ through the required order.
3. For every input block $t^n P_n(L)$ with $n \leq N$, compute $t^n R^n P_n(L + V)$.
4. Sum and truncate after each multiplication.

If negative n occur, R^n is an ordinary positive power for $-n > 0$. Guard terms are needed because an initial t^{-1} block can shift a term of R^n one place toward the retained range.

Input: $F = \sum_{n=n_0}^N P_n t^n$, $G = X + \Delta$, target N

Output: $F \circ G \pmod{t^{N+1}}$

$U \leftarrow t\Delta$;

$R \leftarrow (1 + U)^{-1} \pmod{t^{N-n_0+1}}$;

$V \leftarrow \log(1 + U) \pmod{t^{N-n_0+1}}$;

$H \leftarrow 0$;

for $n \leftarrow n_0$ **to** N **do**

$H \leftarrow H + t^n R^n P_n(L + V)$;

$H \leftarrow H \pmod{t^{N+1}}$;

end

return H ;

Algorithm 2: Near-identity composition through outer block N

Pitfall

Never substitute $L \mapsto \log G$ while leaving t unchanged. The symbols t and L are linked by $t = X^{-1}$ and $L = \log X$; composition must transform both. Updating only one generator is a common source of plausible but false coefficients.

Summary

For $G = X(1 + U)$ with $U \prec 1$, composition is exact generator substitution: $t \mapsto t(1 + U)^{-1}$ and $L \mapsto L + \log(1 + U)$. This construction is coefficientwise finite, respects algebraic operations, and makes associativity transparent.

Chapter 8

Taylor composition and low-order coefficient laws

8.1 The formal Taylor formula

Let $G = X + \Delta$ with $\Delta \in \mathcal{C}_L[[t]]$, so $t\Delta \prec 1$. Since differentiation raises outer order, the formal Taylor family

$$F(X + \Delta) = \sum_{r \geq 0} \frac{\Delta^r}{r!} \mathcal{D}^r F(X) \quad (8.1)$$

is coefficientwise summable.

Theorem 8.1 (Taylor composition in the one-logarithm field). *For $F \in \widehat{\mathcal{PL}}$ and $G = X + \Delta$ as above,*

$$F \circ G = \sum_{r \geq 0} \frac{\Delta^r}{r!} \mathcal{D}^r F. \quad (8.2)$$

At any fixed outer order N , only finitely many r contribute.

Proof. For a block $P(L)t^n$, [Theorem 6.2](#) shows that $\mathcal{D}^r(Pt^n)$ has outer order $n + r$. Since $\text{ord}_t \Delta \geq 0$, the r th Taylor term has order at least $n + r$, so only $r \leq N - n$ can affect block N . It remains to identify the sum with generator substitution. Applied to $t = X^{-1}$, the Taylor series is

$$\sum_{r \geq 0} \frac{\Delta^r}{r!} \mathcal{D}^r t = t \sum_{r \geq 0} (-t\Delta)^r = \frac{t}{1 + t\Delta} = \frac{1}{G}.$$

Applied to $L = \log X$, it is

$$L + \sum_{r \geq 1} \frac{(-1)^{r-1}}{r} (t\Delta)^r = L + \log(1 + t\Delta) = \log G.$$

The equality then extends to polynomials, products, inverses, and summable series. $\square$

8.2 When Taylor's formula is not justified

The condition is not merely that Δ tends to zero numerically. What is needed is that successive differentiated terms decrease in the transseries ordering. In a larger exponential field, take $F(X) = \exp(\exp X)$ and $\Delta = \exp(-X/2)$. Although $\Delta \rightarrow 0$,

$$\Delta F'(X) = \exp(-X/2) \exp X \exp(\exp X) = \exp(X/2) F(X),$$

which is larger than $F(X)$. Thus the naive Taylor hierarchy fails. The polynomial-logarithmic near-identity class avoids this problem because each derivative supplies a factor X^{-1} that dominates polynomial logarithmic growth.

8.3 Low-order composition formulas

Write

$$F(X) = X + f_0(L) + tf_1(L) + t^2f_2(L) + t^3f_3(L) + \mathcal{O}_t(t^4), \quad (8.3)$$

$$G(X) = X + g_0(L) + tg_1(L) + t^2g_2(L) + t^3g_3(L) + \mathcal{O}_t(t^4). \quad (8.4)$$

Let

$$H = F \circ G = X + h_0 + th_1 + t^2h_2 + t^3h_3 + \mathcal{O}_t(t^4).$$

Primes below mean derivatives with respect to L .

Proposition 8.2 (Composition through three inverse-power blocks). *The coefficients are*

$$h_0 = f_0 + g_0, \quad (8.5)$$

$$h_1 = g_1 + f_1 + g_0f'_0, \quad (8.6)$$

$$h_2 = g_2 + f_2 + g_1f'_0 + g_0(f'_1 - f_1) + \frac{g_0^2}{2}(f''_0 - f'_0), \quad (8.7)$$

$$\begin{aligned} h_3 = & g_3 + f_3 + g_2f'_0 + g_1(f'_1 - f_1) + g_0(f'_2 - 2f_2) \\ & + g_0g_1(f''_0 - f'_0) + \frac{g_0^2}{2}(f'''_1 - 3f'_1 + 2f_1) \\ & + \frac{g_0^3}{6}(f''''_0 - 3f''_0 + 2f'_0). \end{aligned} \quad (8.8)$$

Proof. Set $U = t(G - X)$. Through t^3 ,

$$\begin{aligned} U &= g_0t + g_1t^2 + g_2t^3 + \cdots, \\ \log(1 + U) &= g_0t + \left(g_1 - \frac{1}{2}g_0^2\right)t^2 + \left(g_2 - g_0g_1 + \frac{1}{3}g_0^3\right)t^3 + \cdots, \\ (1 + U)^{-1} &= 1 - g_0t + (g_0^2 - g_1)t^2 + \cdots, \\ (1 + U)^{-2} &= 1 - 2g_0t + \cdots. \end{aligned}$$

Insert these in (7.4), Taylor-expand the f_j in the logarithmic increment, and collect blocks. The displayed formulas result. $\square$

8.4 A check using a simple shift

If $G = X + c$ with constant c , then $g_0 = c$ and all later g_j vanish. The formulas reduce to

$$\begin{aligned} h_0 &= f_0 + c, \\ h_1 &= f_1 + cf'_0, \\ h_2 &= f_2 + c(f'_1 - f_1) + \frac{c^2}{2}(f''_0 - f'_0), \end{aligned}$$

which agrees with the ordinary Taylor expansion in c because

$$\mathcal{D}f_0(L) = tf'_0, \quad \mathcal{D}^2f_0(L) = t^2(f''_0 - f'_0).$$

8.5 Noncommutativity

Composition is associative but not commutative. Let

$$F = X + L, \quad G = X + L^2.$$

Then

$$\begin{aligned} F \circ G &= X + L^2 + L + L^2 t - \frac{1}{2} L^4 t^2 + \dots, \\ G \circ F &= X + L + L^2 + 2L^2 t + (L^2 - L^3) t^2 + \dots. \end{aligned}$$

Therefore

$$F \circ G - G \circ F = -L^2 t + \mathcal{O}_t(t^2). \quad (8.9)$$

At first order, the commutator is governed by the vector-field bracket $f_0 g'_0 - g_0 f'_0$. This is the beginning of the Lie-theoretic structure behind formal iteration.

8.6 Taylor form versus generator form

The generator formula is often best when F is explicitly stored as blocks $P_n(L)t^n$. The Taylor formula is best when derivatives of F are already available or when one seeks universal coefficient identities. They have the same formal content, but different computational profiles:

	Generator substitution	Taylor composition
Core data	$(1 + U)^{-1}$ and $\log(1 + U)$	$\mathcal{D}^r F$ and $\Delta^r / r!$
Best for	sparse coefficient blocks	symbolic universal formulas
Truncation	powers of U	derivative order
Main risk	forgetting one generator	using an invalid smallness condition

Summary

Near the identity, formal Taylor composition is a theorem because every derivative raises the inverse-power order. It is equivalent to exact substitution of $1/X$ and $\log X$. The first coefficient laws are triangular and expose both noncommutativity and the mechanism needed for series reversal.

Chapter 9

The near-identity composition group

9.1 The group of tangent-to-identity transseries at infinity

Define

$$\mathcal{G}_{\mathcal{D}\mathcal{L}} = \{F = X + A(X) : A \in \mathcal{C}_L[[t]]\}. \quad (9.1)$$

Every such F satisfies $F/X = 1 + tA = 1 + o(1)$ formally. The identity is X . Closure under composition follows from the preceding chapters.

Theorem 9.1 (Near-identity group). *The set $\mathcal{G}_{\mathcal{D}\mathcal{L}}$ is a group under composition. Every $F = X + A$ has a unique inverse of the form $X + B$ with $B \in \mathcal{C}_L[[t]]$.*

The proof will be constructive in Part IV: fixed-point iteration gains one outer block at each step. The result is a concrete one-logarithm instance of the compositional group of large positive transseries treated by Edgar [4].

9.2 Filtration by contact order

For $F = X + A$ and $G = X + B$, say that they agree through contact order r when $A - B = \mathcal{O}_t(t^r)$. If $H = X + C$ is fixed, then

$$F - G = \mathcal{O}_t(t^r) \implies H \circ F - H \circ G = \mathcal{O}_t(t^r), \quad (9.2)$$

and, because $H' = 1 + \mathcal{O}_t(t)$,

$$F \circ H - G \circ H = \mathcal{O}_t(t^r). \quad (9.3)$$

This filtration makes composition continuous and allows infinite coefficient constructions by stabilization.

9.3 The first commutator block

Let $F = X + f(L)$ and $G = X + g(L)$ with no inverse-power corrections. Using (8.6),

$$\begin{aligned} F \circ G &= X + f + g + t g f' + \mathcal{O}_t(t^2), \\ G \circ F &= X + f + g + t f g' + \mathcal{O}_t(t^2). \end{aligned}$$

Hence

$$F \circ G - G \circ F = t(gf' - fg') + \mathcal{O}_t(t^2). \quad (9.4)$$

The bilinear operation $[f, g] = gf' - fg'$ is the usual bracket of one-dimensional vector fields after the change of independent variable $L = \log X$.

9.4 Iterates and fractional iteration

Integer iterates $F^{[n]}$ are defined by repeated composition. Because $\mathcal{G}_{\mathcal{P}\mathcal{L}}$ is filtered and pro-unipotent at finite truncation, one can formally introduce the substitution operator

$$\mathcal{S}_F : H \mapsto H \circ F$$

and, when $\mathcal{S}_F - I$ raises the filtration, define

$$\log \mathcal{S}_F = \sum_{r \geq 1} \frac{(-1)^{r+1}}{r} (\mathcal{S}_F - I)^r. \quad (9.5)$$

Then a formal fractional iterate may be written

$$\mathcal{S}_F^s = \exp(s \log \mathcal{S}_F), \quad F^{[s]} = X \mathcal{S}_F^s.$$

Developing this construction fully leads to iterative logarithms and Abel coordinates, beyond the central scope here; Edgar’s work on fractional iteration provides a transseries treatment [5]. The key point for the present article is that the same filtration that legitimizes composition also legitimizes recursive and operator-logarithmic constructions.

9.5 Coefficient stabilization under iteration

Suppose a map Φ satisfies

$$U - V = \mathcal{O}_t(t^r) \implies \Phi(U) - \Phi(V) = \mathcal{O}_t(t^{r+1}). \quad (9.6)$$

Then the iterates $U_{n+1} = \Phi(U_n)$ stabilize one additional block per step: $U_{n+1} - U_n = \mathcal{O}_t(t^n)$ after a suitable initialization. This is the formal analogue of a contraction mapping, but no numerical norm is needed. The ordered filtration supplies the notion of increasing precision.

Summary

Near-identity polynomial–logarithmic transseries form a noncommutative group. Their t -adic contact filtration makes composition continuous, identifies the first commutator with a vector-field bracket, and supports fixed points, inverses, and even formal fractional iteration.

Chapter 10

Scale-changing composition and deeper logarithms

10.1 A general leading inner scale

Suppose

$$G(X) = cX^\rho L^\sigma(1 + S), \quad c \neq 0, \quad \rho > 0, \quad S \prec 1. \quad (10.1)$$

On a real positive ray take $c > 0$; in the complex plane choose branches. Then

$$\frac{1}{G} = c^{-1}t^\rho L^{-\sigma}(1 + S)^{-1}, \quad (10.2)$$

$$\log G = \rho L + \sigma \log L + \log c + \log(1 + S). \quad (10.3)$$

These two formulas completely determine $F \circ G$.

10.2 What new scales are forced?

Equation (10.2) may require fractional powers t^ρ . Equation (10.3) introduces $\log L$ if $\sigma \neq 0$. If the coefficients of F already involve $\log L$, composition may create $\log(\rho L + \sigma \log L + \dots)$ and hence another nested expansion. The correct ambient field is chosen by examining these transformed generators, not by guessing from the appearance of F alone.

Example 10.1 (A scale-changing substitution). Let

$$F(X) = X + \log X + X^{-1}, \quad G(X) = 3X^2L^{-1}.$$

Then

$$\begin{aligned} F \circ G &= 3X^2L^{-1} + \log(3X^2L^{-1}) + \frac{L}{3X^2} \\ &= 3X^2L^{-1} + 2L - \log L + \log 3 + \frac{1}{3}Lt^2. \end{aligned}$$

A second logarithmic level appears exactly, not merely as a remainder.

10.3 Composition with a logarithm

If $G = \log X = L$, then

$$t \mapsto L^{-1}, \quad L \mapsto \log L.$$

Thus

$$\left(\sum_n P_n(L) t^n \right) \circ \log = \sum_n P_n(\log L) L^{-n}.$$

This lowers the exponential height but increases logarithmic depth. Repeated composition with $\log$ produces the hierarchy $L_1, L_2, L_3, \dots$

10.4 Composition with a power

For $G = X^\rho$ with $\rho > 0$,

$$t \mapsto t^\rho, \quad L \mapsto \rho L.$$

Therefore

$$F(X^\rho) = \sum_n t^{\rho n} P_n(\rho L). \quad (10.4)$$

This operation stays within an integer lattice only when ρ maps the chosen exponent group into itself. A rational $\rho = p/q$ is handled by the generator $X^{-1/q}$.

10.5 Failure when the inner argument is not large

The formal logarithm $\log G$ requires a leading monomial and a normalized small remainder. If G oscillates across zero or has incomparable dominant terms in an analytic problem, a single branch may not exist on the domain of interest. Likewise, if G is small rather than large, composing an expansion valid at infinity with G has no asymptotic meaning. Formal syntax cannot replace domain and branch analysis.

10.6 An admissibility checklist

Before computing $F \circ G$, determine:

1. Is G in the asymptotic regime for which F is expanded?
2. What is the leading monomial of G ?
3. Can G be factored as leading monomial times $1 + S$ with $S \prec 1$?
4. What do $1/G$ and $\log G$ look like?
5. Do they remain in the chosen exponent lattice and logarithmic depth?
6. In the complex plane, which branches of powers and logarithms are used?
7. Is the resulting family of supports summable at the desired cutoff?

Summary

Scale-changing composition is controlled by the images of the generators. For $G \sim cX^\rho L^\sigma$, one has $1/G \sim c^{-1}X^{-\rho}L^{-\sigma}$ and $\log G = \rho L + \sigma \log L + \log c + o(1)$. Fractional powers and deeper logarithms are therefore often mathematically unavoidable.

Part IV

Series reversal and asymptotic inversion

Chapter 11

Fixed-point construction of the compositional inverse

11.1 The inverse equation

Let

$$F(X) = X + A(X), \quad A \in \mathcal{C}_L[[t]]. \quad (11.1)$$

We seek $G = F^{(-1)}$ in the form

$$G(Y) = Y + B(Y).$$

The equation $F(G(Y)) = Y$ is equivalent to

$$B(Y) = -A(Y + B(Y)). \quad (11.2)$$

Thus B is a fixed point of

$$\Phi(B) = -A \circ (Y + B). \quad (11.3)$$

The variable is written Y during inversion to separate the input of the inverse from the original variable X ; after the calculation one may rename it X .

11.2 The one-block contraction

The derivative of A has outer order at least one. Therefore changing its argument by a block- r error changes the value only at block $r + 1$ or later.

Lemma 11.1 (Filtered Lipschitz gain). *If $B - C = \mathcal{O}_t(t^r)$, with $B, C \in \mathcal{C}_L[[t]]$, then*

$$A(Y + B) - A(Y + C) = \mathcal{O}_t(t^{r+1}). \quad (11.4)$$

Proof. Taylor-expand around $Y + C$:

$$A(Y + B) - A(Y + C) = \sum_{j \geq 1} \frac{(B - C)^j}{j!} (\mathcal{D}^j A)(Y + C).$$

The derivative $\mathcal{D}^j A$ has outer order at least j . The j th term therefore has order at least $jr + j \geq r + 1$ for $r \geq 0$. $\square$

Theorem 11.2 (Existence and uniqueness by coefficient stabilization). *For every F of the form (11.1), the iteration*

$$B_0 = 0, \quad B_{n+1} = -A(Y + B_n) \quad (11.5)$$

stabilizes coefficientwise. Its limit B is the unique solution of (11.2). Consequently F has a unique near-identity compositional inverse $G = Y + B$.

Proof. The first difference $B_1 - B_0 = -A(Y)$ has outer order at least zero. By Theorem 11.1,

$$B_{n+1} - B_n = \mathcal{O}_t(t^n).$$

Hence each fixed block eventually stops changing, which defines a formal limit B . Composition is continuous in the filtration, so the limit satisfies the fixed-point equation. If B and C were two solutions and their first difference occurred at block r , then $B - C = \Phi(B) - \Phi(C) = \mathcal{O}_t(t^{r+1})$, a contradiction. $\square$

11.3 Why a bad rearrangement may fail

For Lambert inversion one can rearrange the equation in more than one way. If $X = We^W$, then $W = \log X - \log W$, which is contractive after the dominant term is removed. The equivalent equation $Q = e^{-Q} - \log X$ for a correction Q is not contractive at the natural starting point: its exponential can create a term as large as X . Edgar uses this example to emphasize that a formal fixed-point scheme must increase asymptotic precision, not merely be algebraically equivalent [3].

11.4 Stopping after a requested order

To compute through t^N , it is enough to perform $N + 1$ fixed-point steps from $B_0 = 0$, although some blocks may stabilize sooner. After every step one may truncate at t^{N+1} because the next application of A cannot bring a discarded block back to an earlier order.

Input: $F = X + A$, target outer block N

Output: $G = F^{(-1)} \pmod{t^{N+1}}$

$B \leftarrow 0$;

for $j \leftarrow 0$ **to** N **do**

$B \leftarrow -A \circ (Y + B) \pmod{t^{N+1}}$;

end

return $Y + B$;

Algorithm 3: Fixed-point compositional inversion

11.5 Left and right inverse

The fixed-point construction gives $F \circ G = Y$. In a group this already implies $G \circ F = X$, but it is useful to see why directly. Construct a right inverse H of G by the same theorem. Then

$$F = F \circ (G \circ H) = (F \circ G) \circ H = H.$$

Thus $G \circ F = G \circ H = Y$. The inverse is two-sided.

Summary

For $F = X + A$ with $A = o(X)$ in the polynomial-logarithmic scale, the equation $B = -A(Y + B)$ is contractive in outer order because $A' = \mathcal{O}_t(t)$. Simple iteration stabilizes one new block at every step and proves both existence and uniqueness of the compositional inverse.

Chapter 12

Triangular coefficient reversal

12.1 Solving one block at a time

Use the notation

$$\begin{aligned} F(X) &= X + f_0 + t f_1 + t^2 f_2 + t^3 f_3 + \cdots, \\ G(Y) &= Y + g_0 + t g_1 + t^2 g_2 + t^3 g_3 + \cdots, \end{aligned}$$

where all coefficients are functions of $L = \log Y$. Set $F \circ G = Y$ in [Theorem 8.2](#). Each equation is linear in the new unknown g_n , with coefficient one. This is the triangular structure of series reversal.

The first three inverse blocks are

$$g_0 = -f_0, \tag{12.1}$$

$$g_1 = -f_1 - g_0 f'_0 = -f_1 + f_0 f'_0, \tag{12.2}$$

$$g_2 = -f_2 - g_1 f'_0 - g_0(f'_1 - f_1) - \frac{g_0^2}{2}(f''_0 - f'_0). \tag{12.3}$$

The formula for g_3 follows by negating every term of (8.8) except g_3 . At higher orders one normally lets a composition routine generate the known residual and then sets the next inverse block equal to its negative.

12.2 Universal triangular algorithm

Suppose $G_{<n} = Y + \sum_{j=0}^{n-1} g_j t^j$ already satisfies

$$F \circ G_{<n} = Y + r_n t^n + \mathcal{O}_t(t^{n+1}).$$

Adding $g_n t^n$ to G changes $F(G)$ by

$$F'(G_{<n}) g_n t^n + \mathcal{O}_t(t^{n+1}) = g_n t^n + \mathcal{O}_t(t^{n+1}),$$

because $F' = 1 + \mathcal{O}_t(t)$. Hence

$$g_n = -r_n. \tag{12.4}$$

This gives an especially simple implementation: compose the current inverse, read the first nonzero residual block, and cancel it.

Input: $F = X + A$, target outer block N
Output: $G = F^{(-1)} \pmod{t^{N+1}}$
 $G \leftarrow Y$;
for $n \leftarrow 0$ **to** N **do**
 $R \leftarrow F \circ G - Y \pmod{t^{n+1}}$;
 $r_n \leftarrow [t^n]R$;
 $G \leftarrow G - r_n t^n$;
end
return G ;

Algorithm 4: Residual-cancellation series reversal

12.3 A seven-parameter example

Consider

$$F(X) = X + aL + b + (cL + d)t + (eL^2 + fL + g)t^2 + \mathcal{O}_t(t^3). \quad (12.5)$$

The inverse has the form

$$G(Y) = Y + g_0(L) + g_1(L)t + g_2(L)t^2 + \mathcal{O}_t(t^3),$$

with

$$g_0 = -(aL + b), \quad (12.6)$$

$$g_1 = (a^2 - c)L + (ab - d), \quad (12.7)$$

$$\begin{aligned} g_2 = & \left(\frac{a^3}{2} - ac - e \right) L^2 \\ & + \left(-a^3 + a^2b + 2ac - ad - bc - f \right) L \\ & - a^2b + \frac{ab^2}{2} + ad + bc - bd - g. \end{aligned} \quad (12.8)$$

These formulas are useful as unit tests for symbolic implementations: random numerical values of the parameters can be substituted and the composition checked through t^2 .

12.4 A one-parameter perturbation of Lambert inversion

Let

$$F(X) = X + \log X + \frac{q}{X} = X + L + qt. \quad (12.9)$$

Substituting $a = 1$, $b = 0$, $c = 0$, $d = q$, and $e = f = g = 0$ into the preceding formulas gives

$$F^{(-1)}(Y) = Y - L + (L - q)t + \left(\frac{1}{2}L^2 - (q + 1)L + q \right) t^2 + \mathcal{O}_t(t^3). \quad (12.10)$$

At $q = 0$ this begins the Wright omega, hence Lambert- W , expansion developed in Part V.

12.5 Degree growth

If $\deg f_0 = d$, repeated composition can make $\deg g_n$ grow roughly linearly with n . In the Lambert case $f_0 = L$, one has exactly $\deg g_n = n$ for $n \geq 1$. The triangular algorithm

never requires an a priori degree guess: coefficient polynomials emerge from arithmetic. For a preallocated dense representation, however, the degree bound supplied by composition formulas can substantially reduce memory.

Summary

Series reversal is triangular because $F' = 1 + \mathcal{O}_t(t)$. Once an approximate inverse is correct through block $n - 1$, the block- n residual is canceled by subtracting exactly that coefficient from the inverse. Universal low-order formulas and residual cancellation provide complementary symbolic methods.

Chapter 13

Newton iteration for compositional inversion

13.1 The Newton map

For a current approximation $G(Y)$ to $F^{(-1)}(Y)$, define the residual

$$R = F \circ G - Y.$$

Newton's correction is

$$G_{\text{new}} = G - \frac{F \circ G - Y}{F' \circ G}. \quad (13.1)$$

The denominator is a unit because $F' = 1 + \mathcal{O}_t(t)$.

Theorem 13.1 (Precision doubling). *If $R = \mathcal{O}_t(t^m)$, then the residual after (13.1) is at least $\mathcal{O}_t(t^{2m+2})$ when $m \geq 0$. In particular, the number of correct outer blocks approximately doubles at each step.*

Proof. Put $C = -R/(F' \circ G)$. Then $C = \mathcal{O}_t(t^m)$. Taylor expansion gives

$$F \circ (G + C) - Y = R + (F' \circ G)C + \sum_{j \geq 2} \frac{C^j}{j!} (F^{(j)} \circ G).$$

The first two terms cancel. Since $F^{(j)} = A^{(j)}$ has outer order at least j , the j th remaining term has order at least $jm + j$. The minimum occurs at $j = 2$ and is $2m + 2$. $\square$

13.2 Newton applied to $X + \log X$

Let $F(X) = X + \log X$ and start with $G_0 = Y$. Then

$$F(G_0) - Y = L, \quad F'(G_0) = 1 + t.$$

The first Newton iterate is

$$\begin{aligned} G_1 &= Y - \frac{L}{1+t} \\ &= Y - L + Lt - Lt^2 + Lt^3 - \dots \end{aligned}$$

It is already correct through block t^1 . A second Newton step produces the correct expansion through at least t^5 if all intermediate arithmetic is carried with sufficient guard precision. Fixed-point iteration would require one step per newly stabilized block.

13.3 Truncated Newton algorithm

Input: $F = X + A$, target outer block N

Output: $G = F^{(-1)} \pmod{t^{N+1}}$

$G \leftarrow Y; p \leftarrow 0;$

while $p \leq N$ **do**

 choose $q \leftarrow \min(N + 1, 2p + 2);$

$R \leftarrow F \circ G - Y \pmod{t^q};$

$J \leftarrow F' \circ G \pmod{t^q};$

$G \leftarrow G - R/J \pmod{t^q};$

$p \leftarrow q;$

end

return $G;$

Algorithm 5: Newton compositional inversion

The variable p records a conservative residual order. In practice one checks the residual after every update rather than relying solely on a predicted precision.

13.4 Cost comparison

Let $M(N)$ denote the cost of truncated multiplication to order N , and let $C(N)$ denote the cost of composition. A triangular method performs roughly N compositions at growing precision, while Newton performs $O(\log N)$ compositions and divisions at geometrically increasing precision. With fast arithmetic this makes Newton asymptotically superior. At low order, however, the triangular recurrence is simpler, exposes exact coefficient structure, and is easier to debug.

13.5 Hybrid strategy

A practical symbolic system often uses:

1. dominant balance to obtain the leading term;
2. two or three triangular blocks to establish the correct scale and branches;
3. Newton iteration to reach high order;
4. a final residual computation as certification.

The methods are not competitors but stages of one robust workflow.

Pitfall

Newton's formula is only as accurate as its denominator and composition. Computing $F' \circ G$ to the final cutoff but computing the residual at lower precision can silently lose blocks after division. Every Newton step must carry a common precision context with guard terms determined by leading orders.

Summary

Compositional Newton iteration subtracts the residual divided by the composed derivative. Because higher derivatives of the correction A begin at progressively later inverse-power blocks, Newton turns an order- m residual into one of order at least $2m + 2$. It is the method of choice for high-order computation.

Chapter 14

The transserial Lagrange–Bürmann formula

14.1 A closed all-orders formula

Fixed-point and Newton iteration are recursive. Lagrange–Bürmann inversion packages every correction into one explicit family.

Theorem 14.1 (Lagrange–Bürmann reversion at infinity). *Let $F(X) = X + A(X)$ with $A \in \mathcal{C}_L[[t]]$. Then*

$$F^{(-1)}(Y) = Y + \sum_{n \geq 1} \frac{(-1)^n}{n!} \mathcal{D}_Y^{n-1}(A(Y)^n). \quad (14.1)$$

The family is coefficientwise summable in the polynomial–logarithmic filtration: the block t^N receives contributions only from $n \leq N + 1$.

Proof. Introduce a formal parameter z and solve

$$U = z\phi(Y + U), \quad \phi = -A.$$

The ordinary formal Lagrange–Bürmann theorem gives

$$U = \sum_{n \geq 1} \frac{z^n}{n!} \mathcal{D}_Y^{n-1}(\phi(Y)^n). \quad (14.2)$$

For completeness, its coefficient form is

$$[z^n]H(Y + U) = \frac{1}{n} [w^{n-1}]H'(Y + w)\phi(Y + w)^n.$$

This follows by the formal residue substitution $z = w/\phi(Y + w)$; an integration by parts reduces the resulting residue to the displayed expression. Taking $H(u) = u$ gives (14.2). Now set $z = 1$. Since A^n has outer order at least zero and $n - 1$ derivatives raise outer order by at least $n - 1$, only $n \leq N + 1$ can affect block N . Thus evaluation at $z = 1$ is legitimate coefficientwise and yields (14.1). $\square$

14.2 The first universal terms

Expanding (14.1) gives

$$F^{(-1)}(Y) = Y - A + AA' - A(A')^2 - \frac{1}{2}A^2A''$$

$$\begin{aligned}
& + A(A')^3 + \frac{3}{2}A^2A'A'' + \frac{1}{6}A^3A''' \\
& - A(A')^4 - 3A^2(A')^2A'' - \frac{1}{2}A^3(A'')^2 - \frac{2}{3}A^3A'A''' - \frac{1}{24}A^4A^{(4)} + \dots \quad (14.3)
\end{aligned}$$

All functions and derivatives on the right are evaluated at Y . The fourth line follows from $-(1/120)\mathcal{D}^4(A^5)$; its partition coefficients are a specialization of the Bell-polynomial formulas reviewed in Appendix A.

14.3 Lagrange formula for an observable

The more general form is often as useful as the inverse itself.

Theorem 14.2 (Composed observable). *For any differentiable formal transseries H for which the family is summable,*

$$H(F^{(-1)}(Y)) = H(Y) + \sum_{n \geq 1} \frac{(-1)^n}{n!} \mathcal{D}_Y^{n-1}(H'(Y)A(Y)^n). \quad (14.4)$$

Proof. Use the general coefficient form of Lagrange–Bürmann in the proof of Theorem 14.1, with $\phi = -A$. $\square$

For example, taking $H(Y) = \log Y$ immediately gives the expansion of the logarithm of an inverse without first computing and then re-expanding the inverse itself.

14.4 Equivalence with fixed-point iteration

Formula (14.1) solves $G = Y - A(G)$, so uniqueness from Theorem 11.2 shows that it equals the fixed-point limit. Conversely, repeatedly substituting $G = Y - A(G)$ and collecting terms by the number of copies of A reconstructs the rooted-tree combinatorics encoded by Lagrange inversion. Newton iteration converges to the same unique element. Thus the four methods differ only in organization:

Method	Organization of the same inverse
Fixed point	one additional filtration block per substitution
Triangular recurrence	one residual coefficient canceled at a time
Newton	many blocks corrected simultaneously, with precision doubling
Lagrange–Bürmann	explicit sum indexed by the number of perturbation copies

14.5 A direct example

For $F = X + a \log X$, take $A = aL$. Then

$$\begin{aligned}
F^{(-1)}(Y) &= Y - aL + a^2Lt + a^3 \left(\frac{1}{2}L^2 - L \right) t^2 \\
&\quad + a^4 \left(\frac{1}{3}L^3 - \frac{3}{2}L^2 + L \right) t^3 + \mathcal{O}_t(t^4).
\end{aligned}$$

The factor a^{n+1} at block t^n follows from the homogeneity of $\mathcal{D}^n(aL)^{n+1}$. Setting $a = 1$ gives the Wright omega expansion.

Summary

For $F = X + A$, the inverse is $Y + \sum_{n \geq 1} (-1)^n \mathcal{D}^{n-1}(A^n)/n!$. Differentiation raises outer order, so the apparently infinite expression is finite at every requested block. The formula unifies recursive inversion, Bell-polynomial combinatorics, and the all-orders Lambert expansion.

Chapter 15

Dominant balance and inverses beyond the identity scale

15.1 Why dominant balance comes first

The preceding chapters assume $F(X) = X + o(X)$. A general asymptotic inverse may have a different leading scale. The correct workflow is:

1. identify a leading model F_0 that can be inverted;
2. write $X = F_0^{(-1)}(Y)$ times or plus a small correction;
3. transform the equation so the remaining unknown is near the identity;
4. apply the polynomial–logarithmic calculus.

Failure to isolate the leading balance can make a formally equivalent fixed point noncontractive.

15.2 Inverting a power–log monomial

Consider

$$Y = cX^\rho(\log X)^\sigma, \quad c > 0, \quad \rho > 0. \quad (15.1)$$

Put

$$S = \log Y, \quad u = \log X.$$

Then

$$S = \log c + \rho u + \sigma \log u. \quad (15.2)$$

The leading approximation is $u \sim S/\rho$. Reversion in the variable u gives

$$u = \frac{1}{\rho} \left[S - \sigma \log S + \sigma \log \rho - \log c + \frac{\sigma^2 \log S + \sigma \log c - \sigma^2 \log \rho}{S} + \mathcal{O}\left(\frac{(\log S)^2}{S^2}\right) \right]. \quad (15.3)$$

Exponentiating yields

$$X = c^{-1/\rho} \rho^{\sigma/\rho} Y^{1/\rho} S^{-\sigma/\rho} \times \left[1 + \frac{\sigma^2 \log S + \sigma \log c - \sigma^2 \log \rho}{\rho S} + \mathcal{O}\left(\frac{(\log S)^2}{S^2}\right) \right]. \quad (15.4)$$

Further terms are generated recursively in powers of S^{-1} with polynomial coefficients in $\log S$.

Derivation of (15.3). Write

$$u = \rho^{-1} \left(S - \sigma \log S + C + \frac{a \log S + b}{S} + \cdots \right).$$

Then

$$\log u = \log S - \log \rho + \frac{-\sigma \log S + C}{S} + \mathcal{O} \left(\frac{(\log S)^2}{S^2} \right).$$

Substitution into (15.2) gives $C = \sigma \log \rho - \log c$, $a = \sigma^2$, and $b = \sigma \log c - \sigma^2 \log \rho$. $\square$

15.3 An exact affine–logarithmic inverse

The equation

$$Y = X + a \log X + b, \quad a \neq 0, \quad (15.5)$$

has the exact solution

$$X = a W_k \left(\frac{1}{a} \exp \left(\frac{Y - b}{a} \right) \right), \quad (15.6)$$

with the branch k selected by the desired domain. Indeed, exponentiating (15.5) after division by a gives

$$\frac{X}{a} \exp(X/a) = \frac{1}{a} \exp((Y - b)/a).$$

This identity is both a useful check on formal reversion and a source of branch information that a purely formal calculation does not record.

15.4 Power times exponential

For

$$X^\alpha e^{\beta X} = Y, \quad \alpha \beta \neq 0, \quad (15.7)$$

one may take an α th root and obtain

$$X e^{(\beta/\alpha)X} = Y^{1/\alpha}.$$

Thus

$$X = \frac{\alpha}{\beta} W_k \left(\frac{\beta}{\alpha} Y^{1/\alpha} \right). \quad (15.8)$$

The large- Y expansion therefore inherits exactly the polynomial–logarithmic structure of W after scaling.

15.5 Perturbing a power–log leading term

Suppose

$$Y = c X^\rho L^\sigma (1 + E(X)), \quad E \prec 1. \quad (15.9)$$

Taking logarithms gives

$$\log Y = \log c + \rho \log L + \sigma \log L + \log(1 + E).$$

First invert the unperturbed left side by (15.3), then treat $\log(1 + E)$ as a small additive perturbation in the L variable. This reduces the problem to near-identity reversion. It is the general mechanism by which multiplicative asymptotic corrections become additive polynomial–logarithmic corrections after a logarithm.

15.6 Critical points and Puiseux reversal

If $F'(X_0) = 0$, local inversion is not near the identity. When

$$F(X) - F(X_0) = c_m(X - X_0)^m + \cdots, \quad c_m \neq 0,$$

the inverse begins with

$$X - X_0 \sim c_m^{-1/m}(Y - F(X_0))^{1/m}.$$

Fractional powers are unavoidable, and there are generally m branches. Lambert W near $-e^{-1}$ is the important case $m = 2$: its branch-point expansion is Puiseux rather than polynomial-logarithmic. One should not try to force such a local inverse into the large-argument scale.

Summary

A non-near-identity inverse begins with dominant balance. Power-log leading terms are inverted by taking logarithms and reverting an affine-log equation; exact affine-log and power-exponential equations reduce to Lambert W . Critical points instead produce Puiseux branches and require a different local scale.

Part V

Lambert W and the canonical logarithmic inverse

Chapter 16

Wright omega as the inverse of $X + \log X$

16.1 From Lambert W to an additive inverse problem

Lambert's function satisfies

$$W(z)e^{W(z)} = z.$$

Choose a logarithm branch and put

$$L_{1,k} = \operatorname{Log} z + 2\pi i k, \quad L_{2,k} = \operatorname{Log} L_{1,k}. \quad (16.1)$$

Taking logarithms of the defining equation on compatible branches gives

$$W_k(z) + \operatorname{Log} W_k(z) = L_{1,k}. \quad (16.2)$$

Thus $W_k(z)$ is obtained by inverting

$$F(X) = X + \operatorname{Log} X$$

at the large argument $Y = L_{1,k}$.

The Wright omega function packages this additive problem directly:

$$\omega(Y) + \operatorname{Log} \omega(Y) = Y, \quad \omega(Y) = W(e^Y) \quad (16.3)$$

where branches are chosen consistently. The DLMF treats both W and ω and records their branch structure and asymptotic expansions [2]. For formal calculation, (16.3) simply says

$$\omega = (X + \log X)^{\langle -1 \rangle}. \quad (16.4)$$

16.2 The first terms by fixed-point iteration

Equation (16.3) is

$$\omega = Y - \log \omega. \quad (16.5)$$

The dominant term is Y . Starting with $\omega_0 = Y$ gives

$$\omega_1 = Y - L, \quad L = \log Y.$$

Substitute again:

$$\omega_2 = Y - \log(Y - L)$$

$$\begin{aligned}
&= Y - L - \log(1 - L/Y) \\
&= Y - L + \frac{L}{Y} + \frac{L^2}{2Y^2} + \frac{L^3}{3Y^3} + \cdots.
\end{aligned}$$

This is not yet correct at every displayed block because the correction inside the logarithm must itself be iterated. Coefficient stabilization or the closed formula below produces

$$\begin{aligned}
\omega(Y) = Y - L + \frac{L}{Y} + \frac{\frac{1}{2}L^2 - L}{Y^2} + \frac{\frac{1}{3}L^3 - \frac{3}{2}L^2 + L}{Y^3} \\
+ \frac{\frac{1}{4}L^4 - \frac{11}{6}L^3 + 3L^2 - L}{Y^4} + \mathcal{O}\left(\frac{L^5}{Y^5}\right).
\end{aligned} \tag{16.6}$$

The first two scales Y and $L = \log Y$ determine the entire hierarchy.

16.3 Derivation by Lagrange–Bürmann

In [Theorem 14.1](#), take $A(Y) = \log Y = L$. Then

$$\omega(Y) = Y + \sum_{n \geq 1} \frac{(-1)^n}{n!} \mathcal{D}_Y^{n-1} L^n. \tag{16.7}$$

The term indexed by n lies entirely in outer block $Y^{-(n-1)}$. Therefore there is exactly one Lagrange term per inverse-power block—an exceptional simplification that explains the clean all-orders formula.

Writing $r = n - 1$ gives

$$\omega(Y) = Y - L + \sum_{r \geq 1} \frac{P_r(L)}{Y^r}, \tag{16.8}$$

where P_r is a polynomial of degree r . The next chapter develops these polynomials in detail.

16.4 A formal generating equation

Introduce independent formal variables s and z , and define

$$Q(s, z) = \sum_{r \geq 1} P_r(z) s^r. \tag{16.9}$$

Substitute

$$\omega(Y) = Y - z + Q(s, z), \quad s = Y^{-1}, \quad z = \log Y,$$

into $\omega + \log \omega = Y$. Since

$$\log \omega = z + \log(1 - sz + sQ),$$

one obtains the compact implicit equation

$$Q = -\log(1 - sz + sQ). \tag{16.10}$$

Conversely, the unique solution of [\(16.10\)](#) with $Q(0, z) = 0$ generates all Lambert polynomials. This is a two-variable ordinary power series problem whose coefficient ring is $\mathbb{K}[z]$.

16.5 A numerical illustration

Let $Y = 100$, so the corresponding Lambert argument is $z = e^{100}$. The exact value is approximately

$$W(e^{100}) = 95.44148664557583184017204\dots$$

Successive truncations of (16.8) have absolute errors:

Largest retained block	Absolute error
$Y - L$	4.67×10^{-2}
P_1/Y	6.05×10^{-4}
P_2/Y^2	5.27×10^{-6}
P_3/Y^3	8.15×10^{-8}
P_4/Y^4	5.60×10^{-9}
P_5/Y^5	1.54×10^{-10}
P_6/Y^6	2.03×10^{-12}

The table is illustrative rather than an error theorem. A residual-based bound is developed in [Chapter 18](#).

16.6 Why the Lambert expansion is the model example

The expansion displays almost every structural feature of the present calculus in its simplest form:

- a logarithm appears because the defining equation contains an exponential;
- a second logarithm appears when the leading term is substituted back;
- each inverse power of the first logarithm carries a polynomial in the second;
- composition and reversion are essential, not optional notation;
- complex branches are represented by changing $L_{1,k}$;
- a different scale, namely a square-root Puiseux series, takes over at the branch point $-e^{-1}$.

Summary

The Wright omega function is the compositional inverse of $X + \log X$. Lambert $W_k(z)$ is obtained by evaluating this inverse at $L_{1,k} = \text{Log } z + 2\pi ik$. Lagrange inversion contributes exactly one term to each inverse-power block, producing polynomials in $L_{2,k} = \text{Log } L_{1,k}$.

Chapter 17

Lambert polynomials and Stirling numbers

17.1 The differential-operator formula

By [Theorem 6.2](#),

$$\mathcal{D}_Y^r L^{r+1} = Y^{-r} \prod_{j=0}^{r-1} (\partial_L - j) L^{r+1}.$$

Comparing with [\(16.7\)](#) gives the first all-orders formula.

Theorem 17.1 (Differential formula for the Lambert polynomials). *For $r \geq 1$,*

$$P_r(z) = \frac{(-1)^{r+1}}{(r+1)!} \prod_{j=0}^{r-1} (\partial_z - j) z^{r+1}. \quad (17.1)$$

The product is a falling factorial of the differential operator, ∂_z^r .

17.2 The Stirling-number formula

Let $\left[\begin{smallmatrix} r \\ q \end{smallmatrix} \right]$ denote the unsigned Stirling number of the first kind, the number of permutations of r elements with q cycles. The signed falling-factorial identity is

$$\prod_{j=0}^{r-1} (D - j) = \sum_{q=0}^r (-1)^{r-q} \left[\begin{smallmatrix} r \\ q \end{smallmatrix} \right] D^q. \quad (17.2)$$

Applying this to z^{r+1} in [\(17.1\)](#) yields:

Theorem 17.2 (Stirling formula). *For every $r \geq 1$,*

$$P_r(z) = \sum_{m=1}^r \frac{(-1)^{r-m}}{m!} \left[\begin{smallmatrix} r \\ r-m+1 \end{smallmatrix} \right] z^m. \quad (17.3)$$

Consequently,

$$W_k(z) \sim L_{1,k} - L_{2,k} + \sum_{r \geq 1} \sum_{m=1}^r \frac{(-1)^{r-m}}{m!} \left[\begin{smallmatrix} r \\ r-m+1 \end{smallmatrix} \right] \frac{L_{2,k}^m}{L_{1,k}^r}. \quad (17.4)$$

Proof. In (17.2), apply D^q to z^{r+1} and put $m = r + 1 - q$. The factorial $(r + 1)!/m!$ cancels the denominator in (17.1); the signs reduce to $(-1)^{r-m}$. $\square$

Formula (17.4) is the standard Stirling-number expansion recorded by Corless et al. and the DLMF [1, 2].

17.3 A differential and integral recurrence

The Stirling recurrence

$$\begin{bmatrix} r+1 \\ q \end{bmatrix} = r \begin{bmatrix} r \\ q \end{bmatrix} + \begin{bmatrix} r \\ q-1 \end{bmatrix}$$

translates into a simple recurrence for P_r .

Theorem 17.3 (Lambert-polynomial recurrence). *With $P_1(z) = z$,*

$$P'_{r+1}(z) = rP_r(z) - P'_r(z), \quad P_{r+1}(0) = 0. \quad (17.5)$$

Equivalently,

$$P_{r+1}(z) = r \int_0^z P_r(u) du - P_r(z). \quad (17.6)$$

This recurrence computes one polynomial using one integration and one subtraction, with exact rational coefficients.

If $P_r(z) = \sum_m p_{r,m} z^m$, then

$$p_{r+1,m+1} = \frac{r}{m+1} p_{r,m} - p_{r,m+1}, \quad (17.7)$$

where missing coefficients are zero. This is convenient for an array-based implementation.

17.4 The first ten polynomials

The first eight fit comfortably on the page:

$$\begin{aligned} P_1(z) &= z, \\ P_2(z) &= \frac{1}{2}z^2 - z, \\ P_3(z) &= \frac{1}{3}z^3 - \frac{3}{2}z^2 + z, \\ P_4(z) &= \frac{1}{4}z^4 - \frac{11}{6}z^3 + 3z^2 - z, \\ P_5(z) &= \frac{1}{5}z^5 - \frac{25}{12}z^4 + \frac{35}{6}z^3 - 5z^2 + z, \\ P_6(z) &= \frac{1}{6}z^6 - \frac{137}{60}z^5 + \frac{75}{8}z^4 - \frac{85}{6}z^3 + \frac{15}{2}z^2 - z, \\ P_7(z) &= \frac{1}{7}z^7 - \frac{49}{20}z^6 + \frac{203}{15}z^5 - \frac{245}{8}z^4 + \frac{175}{6}z^3 - \frac{21}{2}z^2 + z, \\ P_8(z) &= \frac{1}{8}z^8 - \frac{363}{140}z^7 + \frac{3283}{180}z^6 - \frac{6769}{120}z^5 + \frac{245}{3}z^4 - \frac{161}{3}z^3 + 14z^2 - z. \end{aligned}$$

For reference, the next two are

$$P_9(z) = \frac{1}{9}z^9 - \frac{761}{280}z^8 + \frac{29531}{1260}z^7 - \frac{1869}{20}z^6 + \frac{7483}{40}z^5$$

$$\begin{aligned}
& -189z^4 + 91z^3 - 18z^2 + z, \\
P_{10}(z) = & \frac{1}{10}z^{10} - \frac{7129}{2520}z^9 + \frac{6515}{224}z^8 - \frac{9046}{63}z^7 + \frac{5985}{16}z^6 \\
& - \frac{21091}{40}z^5 + \frac{1575}{4}z^4 - 145z^3 + \frac{45}{2}z^2 - z.
\end{aligned}$$

17.5 Structural properties

The formulas immediately imply:

1. $P_r(0) = 0$.
2. $\deg P_r = r$ and the leading coefficient is $1/r$.
3. The linear coefficient is $(-1)^{r+1}$.
4. The nonzero coefficients alternate in sign when read from highest to lowest degree.
5. All coefficients are rational, and multiplying the coefficient of z^m by $m!$ gives an integer Stirling number up to sign.

The highest-degree term alone gives

$$\frac{P_r(L_2)}{L_1^r} = \frac{1}{r} \left(\frac{L_2}{L_1} \right)^r + \text{lower powers of } L_2.$$

This explains why the natural small ratio is L_2/L_1 .

17.6 Independent coefficient checks

A high-order computation has several inexpensive checks:

1. verify the recurrence (17.5);
2. verify $P_r(0) = 0$, degree r , and leading coefficient $1/r$;
3. compare coefficients with (17.3);
4. substitute the truncated W expansion into $W + \log W = L_1$ and check that the residual begins one block later.

Agreement of these structurally different tests is much stronger than a single numerical comparison.

Summary

The coefficient of L_1^{-r} in Lambert's large-argument expansion is a degree- r polynomial $P_r(L_2)$. It has equivalent differential-operator, Stirling-number, differential-recurrence, and integral-recurrence representations. These formulas give exact all-orders generation and strong internal consistency checks.

Chapter 18

Branches, convergence, and error control for Lambert W

18.1 The branch-indexed expansion

For a fixed integer k , set $L_{1,k}$ and $L_{2,k}$ as in (16.1), with compatible logarithm branches. Then, as $|L_{1,k}| \rightarrow \infty$ in an admissible sector,

$$W_k(z) \sim L_{1,k} - L_{2,k} + \sum_{r \geq 1} \frac{P_r(L_{2,k})}{L_{1,k}^r}. \quad (18.1)$$

The branch number enters first through the additive term $2\pi ik$ in $L_{1,k}$; the rest of the coefficient machinery is universal. Corless et al. give the classical branch analysis and all-branch expansion [1]; the DLMF records the current standard notation and branch cuts [2].

18.2 The principal real branch at infinity

For $x > 0$, take

$$L_1 = \log x, \quad L_2 = \log \log x.$$

Then

$$\begin{aligned} W_0(x) = L_1 - L_2 + \frac{L_2}{L_1} + \frac{L_2^2 - 2L_2}{2L_1^2} \\ + \frac{2L_2^3 - 9L_2^2 + 6L_2}{6L_1^3} + \frac{3L_2^4 - 22L_2^3 + 36L_2^2 - 12L_2}{12L_1^4} + \dots \end{aligned} \quad (18.2)$$

Every truncation is real once $x > 1$ and L_2 is defined. For numerical work not very far from the branch point, however, a local series or Newton method is usually preferable to a large-argument expansion.

18.3 The lower real branch near zero

Let $0 < \varepsilon \rightarrow 0$ and write

$$W_{-1}(-\varepsilon) = -u, \quad u > 0.$$

The equation $We^W = -\varepsilon$ becomes

$$ue^{-u} = \varepsilon.$$

With

$$S = \log(1/\varepsilon), \quad L = \log S,$$

we have

$$u - \log u = S. \quad (18.3)$$

This is the inverse of $X - \log X$. Applying [Theorem 14.1](#) with $A = -\log X$ gives

$$u = S + L + \sum_{r \geq 1} \frac{(-1)^{r+1} P_r(L)}{S^r}. \quad (18.4)$$

Therefore

$$\begin{aligned} W_{-1}(-\varepsilon) = & -S - L - \frac{L}{S} - \frac{L - \frac{1}{2}L^2}{S^2} \\ & - \frac{L - \frac{3}{2}L^2 + \frac{1}{3}L^3}{S^3} - \frac{L - 3L^2 + \frac{11}{6}L^3 - \frac{1}{4}L^4}{S^4} + \dots \end{aligned} \quad (18.5)$$

The same Lambert polynomials reappear with alternating block signs.

18.4 The branch point uses a different scale

At $z = -e^{-1}$, the derivative of $w \mapsto we^w$ vanishes at $w = -1$. Put

$$p = \sqrt{2(1 + ez)}.$$

On the principal side,

$$W_0(z) = -1 + p - \frac{1}{3}p^2 + \frac{11}{72}p^3 - \frac{43}{540}p^4 + \frac{769}{17280}p^5 + \dots, \quad (18.6)$$

while the adjacent branch changes the signs of the odd powers. This is a Puiseux expansion in a square root. It is not a failure of polynomial-logarithmic asymptotics; it is a different local geometry caused by a critical point.

18.5 Formal asymptotics versus convergence

Equation (16.10) and the analytic implicit-function theorem show that, for each fixed z , the series $Q(s, z)$ converges for sufficiently small $|s|$. Along the Lambert path $s = 1/L_1$, $z = L_2$, the quantity $|sz| = |L_2/L_1|$ tends to zero. Hence the series is genuinely convergent in a sufficiently far large-argument region on a fixed branch. Outside a convergence region, its finite truncations still provide the standard Poincaré asymptotic expansion in admissible sectors. Precise domains and branch qualifications are given in the classical Lambert- W literature [1, 2].

18.6 Residual-based error estimation

Let

$$w_N = L_1 - L_2 + \sum_{r=1}^N \frac{P_r(L_2)}{L_1^r} \quad (18.7)$$

and define the additive residual

$$R_N = w_N + \text{Log } w_N - L_1. \quad (18.8)$$

The exact solution w satisfies $F(w) = L_1$ for $F(u) = u + \text{Log } u$. Since $F'(u) = 1 + 1/u$, one Newton correction is

$$\delta_N = -\frac{R_N}{1 + 1/w_N}. \quad (18.9)$$

Taylor's theorem gives

$$w - w_N = \delta_N + \mathcal{O}\left(\frac{R_N^2}{|w_N|^2}\right) \quad (18.10)$$

provided the segment or complex neighborhood stays away from 0 and the chosen branch cut. Thus the residual supplies both a check and an improved approximation.

For the formal truncation,

$$R_N = \mathcal{O}\left(\frac{L_2^{N+1}}{L_1^{N+1}}\right),$$

so the inverse error has the same leading order because $F'(w) = 1 + o(1)$. This is the analytic reflection of the triangular residual-cancellation algorithm.

18.7 Branch bookkeeping rules

In complex calculations record all of the following:

1. the branch of $\text{Log } z$ used to form $L_{1,k}$;
2. the integer k ;
3. the branch of $\text{Log } L_{1,k}$ used for $L_{2,k}$;
4. the sector in which $|L_{1,k}| \rightarrow \infty$;
5. any continuation path crossing or avoiding branch cuts.

Two algebraically identical-looking expansions can represent different branches if these data differ.

Summary

The same Lambert polynomials govern every large complex branch through $L_{1,k} = \text{Log } z + 2\pi ik$. The lower real branch near zero is the inverse of $X - \log X$ and uses alternating polynomial blocks. Near $-e^{-1}$ the scale changes to a square-root Puiseux series. Residual evaluation turns a formal truncation into a practical error estimator and Newton correction.

Chapter 19

Generalized Lambert-type inversions and applications

19.1 Equations reducible exactly to W

A surprising number of inverse problems reduce to W after elementary normalization.

19.1.1 Power times exponential

As shown in (15.8),

$$X^\alpha e^{\beta X} = Y \implies X = \frac{\alpha}{\beta} W_k \left(\frac{\beta}{\alpha} Y^{1/\alpha} \right).$$

Its large- Y expansion follows by scaling L_1 and L_2 .

19.1.2 A linear term plus a logarithm

Equation $Y = X + a \log X + b$ has the exact inverse

$$X = a W_k \left(a^{-1} e^{(Y-b)/a} \right).$$

Expanding this exact expression reproduces the direct reversion coefficients from [Chapter 12](#).

19.1.3 The inverse of $X \log X$

If

$$Y = X \log X,$$

put $u = \log X$. Then $ue^u = Y$, so

$$X = e^{W(Y)} = \frac{Y}{W(Y)}. \tag{19.1}$$

With $L_1 = \log Y$ and $L_2 = \log L_1$, division of the Lambert expansion gives

$$X = \frac{Y}{L_1} \left[1 + \frac{L_2}{L_1} + \frac{L_2^2 - L_2}{L_1^2} + \frac{L_2^3 - \frac{5}{2}L_2^2 + L_2}{L_1^3} + \mathcal{O}\left(\frac{L_2^4}{L_1^4}\right) \right]. \tag{19.2}$$

This is a compact example in which reversion is followed by formal division.

19.2 Perturbed affine-log equations

Consider

$$Y = X + a \log X + b + \frac{c \log X + d}{X} + \frac{e(\log X)^2 + f \log X + g}{X^2} + \dots \quad (19.3)$$

One may either:

1. use the exact affine-log inverse as a new leading variable, then compose the perturbation around it; or
2. regard the entire right side as $X + A(X)$ and apply direct near-identity reversion.

The first approach can reduce intermediate coefficient size when a is not small; the second is simpler and produces the universal formulas (12.6)–(12.8) immediately.

19.3 Normal tail and inverse-error-function scales

Let $p \rightarrow 0^+$ and let q be the upper standard-normal quantile, $\Pr(Z > q) = p$. The classical tail expansion has the form

$$p = \frac{e^{-q^2/2}}{\sqrt{2\pi}q} \left(1 - q^{-2} + 3q^{-4} - 15q^{-6} + \dots \right).$$

Taking logarithms and putting $S = \log(1/p)$ turns the inversion into

$$2S = q^2 + 2 \log q + \log(2\pi) + \text{inverse powers of } q^2.$$

With $Y = q^2$, this is an affine-log equation plus a polynomial-logarithmic perturbation. The first result is

$$q = \sqrt{2S} - \frac{\log S + \log(4\pi)}{2\sqrt{2S}} + \mathcal{O}\left(\frac{(\log S)^2}{S^{3/2}}\right). \quad (19.4)$$

All later terms can be obtained by the same reversion algorithms. This explains why logarithms of logarithms occur in extreme quantiles.

19.4 Formal inversion of the logarithmic integral

The logarithmic integral has the large- x expansion

$$\text{li}(x) \sim \frac{x}{\log x} \left(1 + \frac{1}{\log x} + \frac{2!}{(\log x)^2} + \frac{3!}{(\log x)^3} + \dots \right). \quad (19.5)$$

Its inverse therefore belongs to the same power-logarithmic reversion class. Writing $L = \log n$ and $\ell = \log L$, formal inversion gives the Cipolla-type expansion

$$x \sim n \left[L + \ell - 1 + \frac{\ell - 2}{L} - \frac{\ell^2 - 6\ell + 11}{2L^2} + \frac{\ell^3 - 9\ell^2 + 23\ell - 11}{6L^3} + \dots \right]. \quad (19.6)$$

When x is specialized to the n th prime, analytic number theory is needed to transfer the formal inverse of li to p_n ; the algebraic coefficient calculation itself is precisely polynomial-logarithmic series reversal. This example shows that the calculus is not specific to explicitly Lambert-solvable equations.

19.5 Stirling inversion

Suppose N is large and one wants to solve $\Gamma(X + 1) = N$. Stirling’s formula gives

$$\log N = X \log X - X + \frac{1}{2} \log(2\pi X) + \frac{1}{12X} + \cdots.$$

The dominant inverse is controlled by $X \log X$ and hence by Lambert W ; the remaining terms are polynomial–logarithmic perturbations. A robust calculation first uses (19.1) for the leading approximation, then applies Newton or triangular reversion to the Stirling correction.

19.6 A reusable reduction pattern

Many applications fit the sequence

implicit equation $\longrightarrow$ take logarithms $\longrightarrow$ isolate a dominant affine-log map
 $\longrightarrow$ near-identity reversion.

The technique applies to combinatorial saddle points, delay equations, large-deviation thresholds, inverse special functions, and algorithmic complexity equations such as $n \log n = N$. The details differ, but the same four operations recur: multiplication, division, composition, and reversal.

Summary

Lambert W is both a special function and a normal form for asymptotic inversion. Exact reductions handle powers times exponentials, affine-log maps, and $X \log X$. More complicated inverse problems become perturbations of these models after taking logarithms, producing the same polynomial blocks in nested logarithms.

Part VI

Worked calculations, implementation, and analytic meaning

Chapter 20

Worked calculations from beginning to end

The preceding chapters separated the operations in order to prove them cleanly. Actual calculations mix them. This chapter therefore carries out several complete examples, including all truncation decisions and residual checks. Throughout,

$$t = Y^{-1}, \quad L = \log Y,$$

unless another variable is explicitly introduced.

20.1 A product and a quotient in logarithmic blocks

Consider

$$\begin{aligned} A &= 1 + (L + 1)t + (L^2 - L)t^2 + \mathcal{O}(t^3), \\ B &= 1 + (2L - 1)t + (3 - L)t^2 + \mathcal{O}(t^3). \end{aligned}$$

The product is obtained by block convolution:

$$\begin{aligned} AB &= 1 + [(L + 1) + (2L - 1)]t \\ &\quad + [(L^2 - L) + (L + 1)(2L - 1) + (3 - L)]t^2 + \mathcal{O}(t^3) \\ &= 1 + 3Lt + (3L^2 - L + 2)t^2 + \mathcal{O}(t^3). \end{aligned}$$

For the quotient, write

$$\frac{A}{B} = 1 + Q_1t + Q_2t^2 + \mathcal{O}(t^3).$$

The triangular division rule gives

$$Q_1 = (L + 1) - (2L - 1) = 2 - L$$

and

$$\begin{aligned} Q_2 &= (L^2 - L) - (3 - L) - (2L - 1)Q_1 \\ &= 3L^2 - 5L - 1. \end{aligned}$$

Consequently

$$\frac{A}{B} = 1 + (2 - L)t + (3L^2 - 5L - 1)t^2 + \mathcal{O}(t^3). \quad (20.1)$$

A reliable check is multiplication by the divisor. Substituting (20.1) into $B(A/B)$ recovers A through t^2 . This simple residual check is worth automating: it detects sign errors, omitted cross terms, and inadequate internal precision.

20.2 A fully mixed polynomial–logarithmic reversion

Let

$$F(X) = X + 2 \log X + 1 + \frac{3 \log X - 1}{X} + \frac{(\log X)^2 + 2}{X^2}. \quad (20.2)$$

We seek $G = F^{(-1)}$ as $Y \rightarrow +\infty$. Put

$$A(X) = F(X) - X.$$

The Lagrange–Bürmann operator formula from [Chapter 14](#) reads

$$G(Y) = Y - A + \frac{1}{2} \mathcal{D}(A^2) - \frac{1}{6} \mathcal{D}^2(A^3) + \frac{1}{24} \mathcal{D}^3(A^4) + \dots, \quad (20.3)$$

where every occurrence of A on the right is evaluated at Y and

$$\mathcal{D} = t \frac{\partial}{\partial L} - t^2 \frac{\partial}{\partial t}.$$

Only the first four displayed terms can contribute through t^3 . Indeed, $A = \mathcal{O}(L)$ and each application of $\mathcal{D}$ raises the outer t -order by one. A direct calculation gives

$$\begin{aligned} G(Y) = Y - 2L - 1 + \frac{L+3}{Y} + \frac{-3L^2+7L-3}{Y^2} \\ + \frac{-\frac{32}{3}L^3+25L^2-6L-\frac{59}{6}}{Y^3} + \mathcal{O}\left(\frac{L^4}{Y^4}\right). \end{aligned} \quad (20.4)$$

20.2.1 How the coefficient blocks arise

For readers checking the calculation by hand, the contributions in (20.3), truncated after t^3 , are

$$\begin{aligned} -A &= -2L - 1 + (1 - 3L)t - (L^2 + 2)t^2, \\ \frac{1}{2} \mathcal{D}(A^2) &= (4L + 2)t + (-6L^2 + 11L + 2)t^2 \\ &\quad + (-4L^3 - 5L^2 + 9L - 4)t^3, \\ -\frac{1}{6} \mathcal{D}^2(A^3) &= (4L^2 - 4L - 3)t^2 \\ &\quad + (-12L^3 + 46L^2 - 11L - \frac{17}{2})t^3, \\ \frac{1}{24} \mathcal{D}^3(A^4) &= (\frac{16}{3}L^3 - 16L^2 - 4L + \frac{8}{3})t^3. \end{aligned}$$

Their sum is (20.4). Notice that a term which is “higher” in the Lagrange index can still contribute to the same outer t -block. The filtration, not the number of displayed summands, determines the stopping rule.

20.2.2 Residual verification by generator substitution

Set $B(Y) = G(Y) - Y$ and $U = tB$. Since

$$G(Y) = Y(1 + U),$$

composition into (20.2) uses the exact substitutions

$$\frac{1}{G(Y)} = \frac{t}{1+U}, \quad \log G(Y) = L + \log(1+U).$$

Expanding these two expressions through t^4 and substituting into $F(G(Y)) - Y$ gives

$$F(G(Y)) - Y = \left(28L^4 - \frac{211}{3}L^3 - 15L^2 + 91L - \frac{59}{3}\right)t^4 + \mathcal{O}(L^5t^5).$$

Thus every coefficient claimed in (20.4) is certified. The nonzero t^4 residual is expected: it is the first block not included in the inverse.

Computation

For an inverse through Y^{-N} , compute one extra residual block. A zero residual through Y^{-N} proves the retained coefficients; the first nonzero block is also useful for estimating the next correction.

20.3 A nested-logarithmic inverse

Let

$$M = \log L, \quad F(X) = X + a \log X + b \log \log X,$$

where a, b are constants. Define

$$A = aL + bM, \quad p = a + \frac{b}{L}.$$

Then

$$A' = \frac{p}{Y}, \quad A'' = -\frac{a + b/L + b/L^2}{Y^2}.$$

The first three Lagrange–Bürmann terms yield

$$F^{(-1)}(Y) = Y - A + \frac{Ap}{Y} + \frac{1}{Y^2} \left[-Ap^2 + \frac{A^2}{2} \left(a + \frac{b}{L} + \frac{b}{L^2} \right) \right] + \mathcal{O}\left(\frac{L^4}{Y^3}\right). \quad (20.5)$$

This formula illustrates an important closure point. The coefficient of Y^{-2} is no longer a polynomial in L and M alone: negative powers of L appear after differentiation. Hence nested-logarithmic reversion naturally lives in an iterated Laurent coefficient field such as

$$\mathbb{K}((L^{-1}))[M],$$

or in a larger recursively defined transseries field if still deeper logarithms occur.

20.4 Reversion by Newton doubling

The same inverse (20.4) can be obtained with far fewer high-order compositions when many blocks are required. Start with

$$G_0(Y) = Y - 2L - 1,$$

and perform

$$G_{j+1} = G_j - \frac{F(G_j) - Y}{F'(G_j)}. \quad (20.6)$$

If $F(G_j) - Y = \mathcal{O}(t^m)$ and $F'(G_j) = 1 + \mathcal{O}(t)$, the next residual is typically $\mathcal{O}(t^{2m})$. Thus the working precision can be doubled at each stage:

$$t, \quad t^2, \quad t^4, \quad t^8, \quad \dots$$

At every stage all products, quotients, logarithms, and compositions are truncated immediately to the current target. Retaining more terms than the target only increases intermediate expression swell.

20.5 A scale-changing inversion with a Lambert core

Consider

$$Y = X^\rho (\log X)^\sigma \left(1 + \frac{c_1}{\log X} + \frac{c_2}{(\log X)^2} + \mathcal{O}((\log X)^{-3}) \right), \quad \rho > 0. \quad (20.7)$$

Put $Z = \log X$ and $S = \log Y$. Taking logarithms gives

$$S = \rho Z + \sigma \log Z + \frac{c_1}{Z} + \frac{c_2 - \frac{1}{2}c_1^2}{Z^2} + \mathcal{O}(Z^{-3}). \quad (20.8)$$

The unperturbed equation $S = \rho Z + \sigma \log Z$ has the exact inverse

$$Z_0 = \begin{cases} \frac{\sigma}{\rho} W\left(\frac{\rho}{\sigma} \exp\left(\frac{S}{\sigma}\right)\right), & \sigma \neq 0, \\ S/\rho, & \sigma = 0. \end{cases}$$

One may now either revert (20.8) directly in the large variable S , or write $Z = Z_0 + \Delta$ and revert only the inverse-power perturbation. Finally,

$$X = \exp Z.$$

This “logarithm, normalize, Lambert, perturb” workflow avoids forcing a scale-changing problem into a near-identity ansatz too early.

20.6 An implicit equation not exactly Lambert-solvable

Suppose

$$Y = X + \alpha \log X + \beta + \frac{\gamma \log X + \delta}{X} + \mathcal{O}\left(\frac{(\log X)^2}{X^2}\right). \quad (20.9)$$

The triangular formulas of [Chapter 12](#) immediately give

$$X = Y - \alpha L - \beta \quad (20.10)$$

$$+ \frac{(\alpha^2 - \gamma)L + \alpha\beta - \delta}{Y} + \mathcal{O}\left(\frac{L^2}{Y^2}\right). \quad (20.11)$$

This two-line computation is often all that an application needs. The important point is not the particular formula but the workflow:

1. choose the dominant variable so that the map is tangent to the identity;
2. write one polynomial or iterated-Laurent coefficient per outer power;
3. solve coefficient blocks in increasing outer order;
4. compose back and inspect the residual.

Chapter 21

Finite-order algorithms and data structures

Formal existence theorems become useful only when accompanied by coefficientwise algorithms. Algorithmic inversion of exp–log functions by multiseries is developed systematically by Salvy and Shackell [8]. The algorithms in this chapter are deliberately finite: they compute the image of a transseries in a quotient by a prescribed filtration ideal. No machine is asked to store an infinite object.

21.1 Sparse block representation

A truncated one-logarithm transseries

$$T = \sum_{n=n_0}^N P_n(L)t^n + \mathcal{O}(t^{N+1})$$

can be stored as a sparse map

$$n \longmapsto P_n.$$

The outer key n is a rational number if fractional powers are allowed. The value P_n is a polynomial in L for $\mathcal{PL}_{\text{poly}}$, a Laurent series in L^{-1} for $\widehat{\mathcal{PL}}$, or another sparse transseries object for nested logarithms.

Canonicalization consists of three operations:

1. combine equal outer exponents;
2. remove zero coefficient blocks;
3. normalize every coefficient object recursively.

Exact rational arithmetic should be used whenever the input coefficients are exact. Floating-point coefficients hide cancellations and make zero testing unreliable.

21.2 A precision object

A practical implementation needs more than a single integer N . A useful precision descriptor records

$$\mathcal{P} = (n_{\min}, n_{\max}; d_L; \mathcal{P}_{\text{inner}}),$$

where $n_{\max}$ is the largest retained outer exponent, d_L optionally bounds polynomial degree, and $\mathcal{P}_{\text{inner}}$ controls an iterated Laurent coefficient field. The degree bound is not part of

the mathematical definition; it is an implementation safeguard. It should be used only when a theorem or a posteriori residual test guarantees that no discarded high logarithmic degree can feed back into a retained block.

21.3 Multiplication and reciprocal

For

$$A = \sum A_n t^n, \quad B = \sum B_n t^n,$$

the product block is

$$(AB)_r = \sum_{i+j=r} A_i B_j.$$

A sparse implementation iterates only over nonzero pairs. If the supports have s_A and s_B retained blocks, naive multiplication costs $\mathcal{O}(s_A s_B)$ coefficient products; dense consecutive support gives the usual quadratic convolution.

For $B = b_0 + \sum_{n \geq 1} b_n t^n$ with invertible b_0 , its reciprocal $C = \sum c_n t^n$ satisfies

$$c_0 = b_0^{-1}, \quad c_n = -b_0^{-1} \sum_{j=1}^n b_j c_{n-j}. \quad (21.1)$$

The same recurrence works when b_0 is a nonconstant invertible element of the coefficient field.

Input: $B = \sum_{n=0}^N b_n t^n$ with invertible b_0

Output: $C = B^{-1} \bmod t^{N+1}$

$c_0 \leftarrow b_0^{-1};$

for $n \leftarrow 1$ **to** N **do**

$s \leftarrow 0;$

for $j \leftarrow 1$ **to** n **do**

$s \leftarrow s + b_j c_{n-j};$

end

$c_n \leftarrow -b_0^{-1} s;$

end

return $\sum_{n=0}^N c_n t^n;$

Algorithm 6: Truncated reciprocal

21.4 Differentiation as an operator on blocks

The derivative is not obtained by treating t and L as independent:

$$\frac{dt}{dX} = -t^2, \quad \frac{dL}{dX} = t.$$

Consequently

$$\mathcal{D} = t \partial_L - t^2 \partial_t \quad (21.2)$$

and

$$\mathcal{D}(P(L)t^n) = (P'(L) - nP(L))t^{n+1}. \quad (21.3)$$

Repeated differentiation can be applied blockwise by

$$\mathcal{D}^m(P(L)t^n) = t^{n+m} \prod_{j=0}^{m-1} (\partial_L - (n+j)) P(L). \quad (21.4)$$

Formula (21.4) is substantially faster than expanding and differentiating the whole expression at every Lagrange step.

21.5 Composition by the two-generator interface

Let

$$G(Y) = Y(1 + U), \quad \text{ord}_t U > 0.$$

Every one-logarithm block series can be composed into G by the interface

$$t_X \mapsto t_Y(1 + U)^{-1}, \quad L_X \mapsto L_Y + \log(1 + U). \quad (21.5)$$

This separates composition into standard operations: reciprocal, logarithm, polynomial evaluation, powering, multiplication, and summation. The order of evaluation matters for efficiency. Compute $(1 + U)^{-1}$ and $\log(1 + U)$ once, cache their powers, and reuse them for every source block.

Input: $F = \sum_{n=n_0}^N P_n(L)t^n$, $G = Y(1 + U)$, target order N

Output: $F \circ G \bmod t_Y^{N+1}$

$R \leftarrow (1 + U)^{-1} \bmod t_Y^{N+1}$;

$H \leftarrow \log(1 + U) \bmod t_Y^{N+1}$;

$S \leftarrow 0$;

foreach retained block (n, P_n) of F **do**

$S \leftarrow S + t_Y^n R^n P_n(L_Y + H) \pmod{t_Y^{N+1}}$;

end

return S ;

Algorithm 7: Generator-based composition

For negative n , $R^n = (1 + U)^{-n}$ is an ordinary positive power. For rational n , use the binomial series. Terms should be truncated after every multiplication.

21.6 Triangular reversion

For

$$F(X) = X + \sum_{n \geq 0} a_n(\log X)X^{-n}, \quad G(Y) = Y + \sum_{n \geq 0} g_n(\log Y)Y^{-n},$$

the coefficient g_N occurs linearly and for the first time in the Y^{-N} block of $F(G) - Y$. This gives the following robust algorithm.

Input: $F = X + A(X)$, target outer order N

Output: $G = F^{(-1)} \bmod Y^{-N-1}$

$G \leftarrow Y$;

for $n \leftarrow 0$ **to** N **do**

 introduce an unknown coefficient block $g_n(L)Y^{-n}$ in G ;

 compute only the Y^{-n} block of $F(G) - Y$;

 solve that block equation for $g_n(L)$;

 substitute the solution and canonicalize;

end

return G ;

Algorithm 8: Triangular polynomial-logarithmic reversion

The “solve” step is normally just negation because the tangent map is the identity. If the leading derivative is a nonzero coefficient c , divide by c . If the derivative vanishes,

ordinary transseries reversion is not the correct local model; fractional powers or a different dominant balance are required.

21.7 Newton reversion with precision doubling

For high order, Newton iteration

$$G_{\text{new}} = G - \frac{F(G) - Y}{F'(G)}$$

usually outperforms one-block-at-a-time reversal. The implementation should not compute every iterate at full final precision. If the current result is correct through order m , compute the next one only through order $2m$. This makes the number of composition stages logarithmic in the requested order.

A safe schedule is:

$$1, 2, 4, 8, \dots, 2^{\lceil \log_2 N \rceil},$$

with the last stage truncated to N . Before each quotient, normalize the leading block of $F'(G)$ and verify that it is invertible.

21.8 Lagrange–Bürmann as a streaming algorithm

The formal identity

$$G - Y = \sum_{n \geq 1} \frac{(-1)^n}{n!} \mathcal{D}^{n-1}(A^n)$$

does not require construction of all terms at once. Maintain the current power A^n and apply the block operator (21.4). Stop when the valuation lower bound

$$\text{ord}_t(\mathcal{D}^{n-1}(A^n)) > N$$

guarantees that no retained block can occur. This is particularly effective when A has positive outer order. When A begins at order t^0 but grows only logarithmically, the derivative count still forces the n th Lagrange term to begin no earlier than t^{n-1} .

21.9 A compact symbolic skeleton

The following pseudocode emphasizes the architecture rather than a particular computer-algebra system.

Listing 21.1: Sparse outer blocks with recursive coefficients

```
class PLSeries:
    # blocks[n] is the coefficient of t**n
    def __init__(self, blocks, precision):
        self.blocks = canonicalize(blocks, precision)
        self.precision = precision

    def truncate(self, N):
        return PLSeries({n: p for n, p in self.blocks.items()
                        if n <= N}, N)

    def mul(self, other, N):
```



```

    out = {}
    for i, ai in self.blocks.items():
        for j, bj in other.blocks.items():
            if i + j <= N:
                out[i+j] = out.get(i+j, 0) + ai*bj
    return PLSeries(out, N)

def derivative(self, N):
    out = {}
    for n, p in self.blocks.items():
        if n + 1 <= N:
            out[n+1] = diff_L(p) - n*p
    return PLSeries(out, N)

def compose_near_identity(self, U, N):
    R = reciprocal(one() + U, N)
    H = log_series(one() + U, N)
    out = zero()
    for n, p in self.blocks.items():
        term = t_power(n) * power(R, n, N)
        term *= substitute_L(p, L + H)
        out = (out + term).truncate(N)
    return out

def inverse_newton(F, G0, N):
    G, m = G0, 1
    while m < N:
        target = min(2*m, N)
        residual = compose(F, G, target) - identity()
        jacobian = compose(derivative(F), G, target)
        G = (G - divide(residual, jacobian, target)).truncate(target)
        m = target
    return G

```

The recursive operations `diff_L`, `substitute_L`, and `canonicalize` select polynomial, Laurent, or deeper transseries coefficients.

21.10 Internal precision and guard blocks

Composition and division may require source terms beyond the desired output order if leading exponents are negative or if a scale change occurs. Accordingly, a production implementation should determine internal precision from valuations rather than simply requesting the same order from every operand.

For near-identity one-logarithm composition with nonnegative outer corrections, one or two guard blocks are usually enough for a residual check, but “usually” is not a theorem. The rigorous rule is dependency tracing: for each output exponent, enumerate all products and substitutions capable of reaching it. Ratio-set or grid-based transseries implementations formalize this bookkeeping by recording the small generators needed for a calculation; see Edgar’s grid formalism and van der Hoeven’s computational framework [3, 11].

21.11 Complexity and expression swell

At fixed logarithmic degree, dense multiplication and triangular division are quadratic in the number of retained outer blocks. Newton inversion reduces the number of high-cost composition stages, and fast polynomial or series arithmetic can lower asymptotic complexity further. In practice the main obstacle is often coefficient swell rather than operation count: polynomial degrees rise, rational numerators grow, and nested coefficient fields expand.

The most effective controls are:

1. truncate after every primitive operation;
2. factor or collect only at block boundaries;
3. cache $\log(1 + U)$, $(1 + U)^\alpha$, and repeated derivatives;
4. postpone full expansion of coefficient polynomials when possible;
5. verify by a residual rather than repeatedly simplifying two enormous expressions to canonical equality.

Chapter 22

From formal transseries to asymptotic expansions

A formal transseries is an algebraic object. An asymptotic expansion is a statement about a function and a limit. The two viewpoints cooperate, but they are not interchangeable without hypotheses.

22.1 Poincaré meaning in an ordered scale

Let (ϕ_ν) be a totally ordered asymptotic scale, so that $\phi_{\nu+1} = o(\phi_\nu)$ in the chosen enumeration. A function f has the expansion

$$f \sim \sum_{\nu} c_{\nu} \phi_{\nu}$$

when, after every finite truncation at $\nu = N$,

$$f - \sum_{\nu \leq N} c_{\nu} \phi_{\nu} = o(\phi_N).$$

For the polynomial–logarithmic scale, a convenient block formulation is

$$f(X) - \sum_{n=n_0}^N X^{-n} P_n(\log X) = o\left(X^{-N} (\log X)^{d_N}\right) \quad (22.1)$$

after the monomials inside the last block have themselves been ordered. One must specify that inner ordering; the symbol $\mathcal{O}(X^{-N})$ alone does not distinguish $X^{-N} \log^{100} X$ from $X^{-N}/\log X$.

22.2 Uniqueness of coefficients

Proposition 22.1 (Uniqueness in an asymptotic scale). *If the monomials of a polynomial–logarithmic expansion form a strictly decreasing asymptotic scale and two expansions represent the same function to every finite order, then their coefficients agree term by term.*

Proof. Subtract the two expansions. If a first nonzero coefficient existed, divide the difference by its monomial. All later monomials tend to zero relative to it, so the quotient would tend to that nonzero coefficient. But the assumed remainder statement says the same quotient tends to zero, a contradiction. $\square$

This elementary argument is the analytic counterpart of uniqueness of Hahn or grid-based transseries support.

22.3 Transfer under arithmetic

Finite asymptotic expansions are stable under addition and multiplication. Division is stable when the divisor has a nonvanishing leading term. More precisely, suppose

$$f = F_N + r_f, \quad g = G_N + r_g,$$

where the remainders are smaller than the last retained monomials and the leading term of G_N dominates r_g . Then formal multiplication of F_N and G_N , followed by truncation, gives the asymptotic product. Similarly, the geometric expansion

$$\frac{1}{g} = \frac{1}{G_N} \frac{1}{1 + r_g/G_N}$$

justifies the formal reciprocal to every order for which $r_g/G_N \rightarrow 0$.

Differentiation is more delicate: a Poincaré expansion cannot always be differentiated termwise. Sufficient conditions include analyticity with uniform sectorial remainder bounds, or direct derivative estimates on the remainders. Formal differentiation is always defined in the transseries algebra; analytic realization requires this extra regularity.

22.4 Composition needs uniform control

Suppose $f(X)$ has an expansion as $X \rightarrow +\infty$ and $g(Y) \rightarrow +\infty$. Substituting $X = g(Y)$ into each displayed term is not enough. The remainder estimate for f must be uniform over the range traversed by $g(Y)$, and the perturbation of logarithms and powers must remain inside the selected branch or sector.

For a real near-identity map

$$g(Y) = Y(1 + u(Y)), \quad u(Y) \rightarrow 0,$$

these conditions are usually straightforward:

$$\log g = \log Y + \log(1 + u), \quad g^{-\alpha} = Y^{-\alpha}(1 + u)^{-\alpha}.$$

For complex composition, one must additionally keep $1 + u$ away from the chosen logarithmic cut and control arguments uniformly.

22.5 A posteriori inverse error from a residual

The most useful analytic bridge is a residual estimate.

Theorem 22.2 (Residual-to-inverse error). *Let f be continuously differentiable and one-to-one on an interval I . Suppose $x_* = f^{-1}(y)$ and an approximation $x_N \in I$ satisfy*

$$|f'(x)| \geq m > 0$$

for every x between x_N and x_ . Then*

$$|x_N - x_*| \leq \frac{|f(x_N) - y|}{m}. \quad (22.2)$$

Proof. By the mean-value theorem,

$$f(x_N) - f(x_*) = f'(\xi)(x_N - x_*)$$

for some ξ between x_N and x_* . Take absolute values and use $|f'(\xi)| \geq m$. $\square$

For $f(X) = X + \mathcal{O}(\log X)$, one has $f'(X) = 1 + o(1)$ under the relevant differentiability hypotheses, so m may be chosen close to 1 for large X . Therefore the first omitted residual block is also the natural scale of the inverse error. This gives a rigorous reason for the residual checks emphasized throughout the article.

22.6 Newton's correction and residual squaring

Let x_* solve $f(x) = y$, and put

$$x_+ = x - \frac{f(x) - y}{f'(x)}.$$

Taylor's theorem gives

$$f(x_+) - y = \frac{1}{2}f''(\xi)(x_+ - x)^2$$

for a suitable ξ , provided f' stays nonzero. Hence a residual of scale ε is changed to scale ε^2 after normalization by a nonvanishing derivative. The formal valuation-doubling property of Newton reversion is the algebraic shadow of this analytic estimate.

22.7 Convergent, asymptotic, and divergent expansions

Three different situations occur.

1. A block series may converge for sufficiently large X .
2. It may be a nonconvergent but valid Poincaré expansion, with every finite truncation improving the asymptotic order.
3. It may be only a formal candidate until a realization theorem is supplied.

The Lambert- W large-argument expansion is commonly used asymptotically, with branch-dependent complex logarithms; its coefficients and branch structure agree with the standard formulas in Corless et al. and the DLMF [1, 2]. Convergence questions depend on how the expansion is reorganized and on the location in the complex plane. Formal coefficient identities should therefore not be advertised as global convergence statements.

22.8 Sectorial and parameter-uniform expansions

For complex X , write

$$L = \operatorname{Log} X$$

on a fixed branch in a sector

$$|\arg X - \theta_0| \leq \Theta < \pi.$$

A sectorial polynomial-logarithmic expansion requires remainder bounds uniform on each closed subsector. If parameters are present, uniformity must also be specified over a parameter set. Coefficients can change character when a leading parameter vanishes: an invertible block may cease to be invertible, a dominant balance may switch, or two scales may resonate.

A good theorem therefore states:

1. the domain and branch;
2. the parameter range;

3. the ordered scale;
4. the truncation convention;
5. the uniform remainder estimate.

22.9 Turning a formal answer into a theorem

A standard proof strategy for an inverse expansion is:

1. construct a formal truncation G_N ;
2. prove G_N lies in a domain where the original function is one-to-one;
3. estimate the residual $R_N = f(G_N) - Y$;
4. prove a lower bound for $|f'|$ near G_N ;
5. apply [Theorem 22.2](#).

This method often avoids proving convergence of the full formal series. Each finite truncation is justified directly.

Chapter 23

Failure modes, diagnostics, and best practices

Polynomial–logarithmic calculations are forgiving at leading order and unforgiving at high order. The following failure modes account for most incorrect formulas.

23.1 Using the wrong monomial order

For $X \rightarrow +\infty$,

$$X^{-1} \prec (\log X)^{-m} \prec 1 \prec (\log X)^m \prec X$$

for every fixed $m > 0$. Inside an X^{-n} block, however, powers of $\log X$ have their own order. Sorting only by total symbolic degree destroys the asymptotic hierarchy.

23.2 Treating t and L as independent under differentiation

The formal partial derivative ∂_t is not the derivative with respect to X . The correct operator is

$$\mathcal{D} = t\partial_L - t^2\partial_t.$$

Forgetting the $t\partial_L$ term loses all derivatives of logarithmic coefficients; forgetting the $-t^2\partial_t$ term gives the wrong outer power.

23.3 Dividing before normalizing the leading block

A quotient recurrence requires an invertible leading coefficient. First factor the leading monomial and coefficient:

$$B = b_\lambda t^\lambda (1 + R), \quad \text{ord}_t R > 0.$$

Then invert b_λ , t^λ , and $1 + R$ separately. Applying a geometric series to an expression whose “small” part is not actually small is invalid.

23.4 Under-expanding intermediate expressions

To obtain an output through t^N , an internal logarithm, reciprocal, or composition may need terms beyond its visually obvious order. Determine precision by valuation dependencies, not by counting printed terms. A final residual one block deeper is the simplest safeguard.

23.5 Using Taylor composition outside its admissible regime

The formula

$$F(Y + H) = \sum_{k \geq 0} \frac{H^k}{k!} F^{(k)}(Y)$$

is coefficientwise meaningful when derivatives and powers force increasing valuation. If H is comparable with or larger than Y , this is not a near-identity substitution. Change the dominant scale first and use exact generator substitution.

23.6 Forgetting that composition is not commutative

Even when $F = X + o(X)$ and $G = X + o(X)$,

$$F \circ G \neq G \circ F$$

in general. The first discrepancy is governed by a differential commutator such as $A'B - B'A$. This matters in iterated normal forms and fractional iteration [5].

23.7 Applying ordinary power-series reversion at a critical point

If $F'(X_0) = 0$, the inverse is generally not an ordinary integer-power series. A zero of multiplicity m leads to a Puiseux scale $(Y - Y_0)^{1/m}$. The square-root expansion of Lambert W at $-e^{-1}$ is the canonical example. No amount of integer-power coefficient matching repairs a wrong local scale.

23.8 Suppressing branches of logarithms

For complex X , both $\text{Log } X$ and $\text{Log Log } X$ are branch dependent. Changing the branch shifts the first logarithm by $2\pi i k$ and propagates through every later block. A formula for W_k must state which $\log_k z$ is being used.

23.9 Writing ambiguous remainder symbols

The estimate

$$\mathcal{O}(X^{-3})$$

does not contain $X^{-3}(\log X)^{100}$ under the usual bounded-coefficient interpretation. Prefer a scale-aware statement such as

$$\mathcal{O}\left(\frac{(\log X)^d}{X^3}\right)$$

or identify the first omitted monomial explicitly.

23.10 Assuming formal differentiation preserves analytic asymptotics

Termwise differentiation needs control of the differentiated remainder. Oscillatory functions can have a tiny remainder and a large derivative. Use analyticity in a sector, monotonicity plus derivative estimates, or another theorem appropriate to the realization.

23.11 Confusing equality of transseries with equality of functions

Different functions can share the same asymptotic expansion, differing by a term beyond every monomial in the chosen scale, such as e^{-X} relative to a pure power-log scale. Enlarging the transseries field may record that term; staying in the smaller field deliberately forgets it.

23.12 Letting coefficient domains grow unnoticed

Starting with polynomial coefficients in L does not guarantee that every operation remains polynomial. Integration, nested-log differentiation, and some scale changes introduce L^{-1} or $\log L$. State the coefficient ring before claiming closure.

23.13 Premature floating-point evaluation

Numerical substitution into huge logarithmic polynomials can cause catastrophic cancellation. Keep exact coefficients, combine blocks symbolically, and evaluate only after choosing the truncation order. For very large arguments, evaluate in nested form rather than expanding high-degree polynomials into monomials.

23.14 No independent verification

Deriving and simplifying an expression with the same code path is not an independent check. Better diagnostics include:

1. multiply a quotient by its divisor;
2. compose an inverse in both orders;
3. compare Newton and triangular reversion;
4. verify several coefficients numerically at high precision;
5. inspect the first omitted residual block.

23.15 A practical checklist

Before accepting a calculation, answer all of the following.

	Question
1	What is the limiting variable and in which domain or sector?
2	What is the exact ordered monomial scale?
3	Which coefficient ring or iterated field is being used?
4	What is the leading monomial of every divisor?
5	Is each composition near-identity, or has the scale first been changed?
6	Which logarithmic branches are fixed?
7	What target filtration and guard precision are required?
8	Can discarded terms feed a retained block?
9	Does the derivative operator respect $t = X^{-1}$ and $L = \log X$?

- 10 Does the inverse map have nonzero leading derivative?
 - 11 Is the result formal, asymptotic, or convergent?
 - 12 What analytic hypotheses justify composition or differentiation?
 - 13 Has the result been substituted back into the defining equation?
 - 14 Does the first nonzero residual have the predicted order?
 - 15 Have coefficients been kept exact until final evaluation?
-

Summary

The safest philosophy is filtration first, algebra second, simplification third. Choose the scale, prove that every operation is coefficientwise finite, truncate aggressively, and certify the answer by a residual.

Appendix A

Bell polynomials and Faà di Bruno bookkeeping

Bell polynomials organize the derivative combinatorics behind composition, Taylor substitution, and Lagrange inversion.

A.1 Partial exponential Bell polynomials

For integers $n \geq k \geq 0$, define

$$B_{n,k}(x_1, x_2, \dots) = \sum \frac{n!}{j_1! j_2! \dots} \prod_{m \geq 1} \left(\frac{x_m}{m!} \right)^{j_m}, \quad (\text{A.1})$$

where the sum is over all nonnegative integer sequences satisfying

$$\sum_{m \geq 1} j_m = k, \quad \sum_{m \geq 1} m j_m = n.$$

The complete Bell polynomial is

$$B_n(x_1, \dots, x_n) = \sum_{k=0}^n B_{n,k}(x_1, \dots, x_{n-k+1}).$$

The first partial polynomials include

$$\begin{aligned} B_{1,1} &= x_1, \\ B_{2,1} &= x_2, & B_{2,2} &= x_1^2, \\ B_{3,1} &= x_3, & B_{3,2} &= 3x_1x_2, & B_{3,3} &= x_1^3, \\ B_{4,2} &= 4x_1x_3 + 3x_2^2, & B_{4,3} &= 6x_1^2x_2, & B_{4,4} &= x_1^4. \end{aligned}$$

A.2 Faà di Bruno's formula

For sufficiently differentiable functions,

$$\mathcal{D}^n(f \circ g) = \sum_{k=0}^n f^{(k)}(g) B_{n,k}(g', g'', \dots, g^{(n-k+1)}). \quad (\text{A.2})$$

In formal differential algebras this is an algebraic identity. It explains why the n th composition coefficient depends polynomially on derivatives of the source coefficients and on partitions of n .

Proof. Expand

$$f(g+h) = \sum_{k \geq 0} \frac{f^{(k)}(g)}{k!} (g(X+h) - g(X))^k.$$

Now

$$g(X+h) - g(X) = \sum_{m \geq 1} \frac{g^{(m)}(X)}{m!} h^m.$$

The coefficient of $h^n/n!$ in the k th power is exactly $B_{n,k}(g', g'', \dots)$. Comparing coefficients proves (A.2). $\square$

A.3 Bell-polynomial form of implicit coefficients

Suppose $G = Y + H$ and $F = X + A$. The inverse equation is

$$H = -A(Y + H).$$

Taylor expansion gives

$$H = - \sum_{k \geq 0} \frac{H^k}{k!} A^{(k)}(Y).$$

Equating outer blocks recursively produces polynomials in $A, A', A'', \dots$. The first universal terms are

$$\begin{aligned} H = & -A + AA' - A(A')^2 - \frac{1}{2}A^2A'' \\ & + A(A')^3 + \frac{3}{2}A^2A'A'' + \frac{1}{6}A^3A''' + \dots \end{aligned} \quad (\text{A.3})$$

These are the expansion of

$$\sum_{n \geq 1} \frac{(-1)^n}{n!} \mathcal{D}^{n-1}(A^n).$$

Bell polynomials provide an efficient way to expand the repeated derivatives without repeatedly applying the product rule from scratch.

A.4 Coefficient extraction with an auxiliary variable

A particularly clean implementation introduces an auxiliary ordinary variable z and uses

$$\exp\left(\sum_{m \geq 1} x_m \frac{z^m}{m!}\right) = \sum_{n \geq 0} B_n(x_1, \dots, x_n) \frac{z^n}{n!}.$$

All Bell polynomials up to degree N can then be generated by a single truncated exponential. This technique is useful when composition coefficients are needed symbolically in terms of unspecified input blocks.

Appendix B

Changing between infinity and zero

Many power-log expansions are written at $x \rightarrow 0^+$ rather than $X \rightarrow +\infty$. The conversion is elementary but sign errors are common.

B.1 The basic dictionary

Set

$$X = x^{-1}, \quad L = \log X = -\log x.$$

Then

$$X^{-n} P(\log X) = x^n P(-\log x). \quad (\text{B.1})$$

Thus

$$\sum_{n \geq n_0} X^{-n} P_n(\log X) \longleftrightarrow \sum_{n \geq n_0} x^n P_n(-\log x).$$

If $\ell = \log L$, then

$$\ell = \log(-\log x)$$

on the positive real axis.

B.2 Derivative conversion

Since $X = x^{-1}$,

$$\frac{d}{dx} = -X^2 \frac{d}{dX}.$$

In the zero-side variables x and $\lambda = -\log x$,

$$\frac{d}{dx} = \partial_x - \frac{1}{x} \partial_\lambda$$

when x and λ are temporarily treated as formal generators. Applying this to $x^n P(\lambda)$ gives

$$\frac{d}{dx}(x^n P(\lambda)) = x^{n-1}(nP(\lambda) - P'(\lambda)),$$

the exact counterpart of (21.3).

B.3 The lower Lambert branch

For $x \rightarrow 0^+$, the solution $W_{-1}(-x)$ tends to $-\infty$. Put

$$S = \log(1/x), \quad M = \log S.$$

Writing $W_{-1}(-x) = -U$ transforms

$$W_{-1}(-x)e^{W_{-1}(-x)} = -x$$

into

$$U - \log U = S.$$

Reversion on the appropriate sign branch gives

$$W_{-1}(-x) = -S - M - \frac{M}{S} + \frac{M^2/2 - M}{S^2} + \mathcal{O}\left(\frac{M^3}{S^3}\right).$$

The same Lambert polynomials occur, but signs and the logarithmic branch are changed by the transformed defining equation.

B.4 Dulac-style power–log series

Near $x = 0^+$, a finite-type power–log series often has the form

$$\sum_{\alpha \in A} x^\alpha P_\alpha(\log x),$$

where A is a well-ordered additive subset of positive reals. Such series appear in local dynamical normal forms and Dulac maps. The power–log literature develops composition, normal forms, and embeddings in this local language [7]. Formula (B.1) shows the direct relation with the large-variable conventions of this article, while the admissible supports may be more general than consecutive integer powers.

Appendix C

Quick-reference formula sheet

This appendix collects the main formulas without proofs.

Variables and scale

$$t = X^{-1}, \quad L = \log X, \quad \mathcal{P}\mathcal{L}_{\text{poly}} = \mathbb{K}[L]((t)), \quad \widehat{\mathcal{P}\mathcal{L}} = \mathbb{K}((L^{-1}))((t)).$$

For $X \rightarrow +\infty$,

$$X^{-1} \prec L^{-m} \prec 1 \prec L^m \prec X \quad (m > 0).$$

Multiplication

If

$$A = \sum A_n t^n, \quad B = \sum B_n t^n,$$

then

$$AB = \sum_r C_r t^r, \quad C_r = \sum_{i+j=r} A_i B_j.$$

Reciprocal and quotient

For $B = \sum_{n \geq 0} b_n t^n$ with invertible b_0 ,

$$c_0 = b_0^{-1}, \quad c_n = -b_0^{-1} \sum_{j=1}^n b_j c_{n-j}, \quad B^{-1} = \sum c_n t^n.$$

For $Q = A/B$,

$$q_n = b_0^{-1} \left(a_n - \sum_{j=1}^n b_j q_{n-j} \right).$$

Powers, logarithm, and exponential

For $\text{ord}_t U > 0$,

$$(1 + U)^\alpha = \sum_{k \geq 0} \binom{\alpha}{k} U^k,$$

$$\log(1+U) = \sum_{k \geq 1} \frac{(-1)^{k+1}}{k} U^k,$$

$$\exp U = \sum_{k \geq 0} \frac{U^k}{k!}.$$

Differentiation

$$\mathcal{D} = t\partial_L - t^2\partial_t,$$

$$\mathcal{D}(P(L)t^n) = (P' - nP)t^{n+1},$$

$$\mathcal{D}^m(P(L)t^n) = t^{n+m} \prod_{j=0}^{m-1} (\partial_L - n - j)P(L).$$

Near-identity composition

For $G(Y) = Y(1+U)$,

$$t_X \mapsto t_Y(1+U)^{-1}, \quad L_X \mapsto L_Y + \log(1+U).$$

Equivalently, for coefficientwise summable Taylor families,

$$F(Y+H) = \sum_{k \geq 0} \frac{H^k}{k!} F^{(k)}(Y).$$

Near-identity reversion

For $F(X) = X + A(X)$,

$$F^{\langle -1 \rangle}(Y) = Y + \sum_{n \geq 1} \frac{(-1)^n}{n!} \mathcal{D}^{n-1}(A(Y)^n). \quad (\text{C.1})$$

The first terms are

$$Y - A + AA' - A(A')^2 - \frac{1}{2}A^2A'' + A(A')^3 + \frac{3}{2}A^2A'A'' + \frac{1}{6}A^3A''' + \dots.$$

Newton iteration is

$$G_{\text{new}} = G - \frac{F(G) - Y}{F'(G)}.$$

Generic affine-log inverse

If

$$F(X) = X + a \log X + b + \frac{c \log X + d}{X} + \mathcal{O}\left(\frac{(\log X)^2}{X^2}\right),$$

then

$$F^{\langle -1 \rangle}(Y) = Y - aL - b + \frac{(a^2 - c)L + ab - d}{Y} + \mathcal{O}\left(\frac{L^2}{Y^2}\right).$$

Lambert and Wright omega

The Wright omega function is the inverse of $X + \log X$:

$$\omega(Y) + \log \omega(Y) = Y, \quad W(z) = \omega(\text{Log } z)$$

with compatible branches. Its expansion is

$$\omega(Y) = Y - L + \sum_{r \geq 1} \frac{P_r(L)}{Y^r}, \quad L = \log Y,$$

where

$$\begin{aligned} P_r(z) &= \frac{(-1)^{r+1}}{(r+1)!} \prod_{j=0}^{r-1} (\partial_z - j) z^{r+1} \\ &= \sum_{m=1}^r \frac{(-1)^{r-m}}{m!} \begin{bmatrix} r \\ r-m+1 \end{bmatrix} z^m. \end{aligned}$$

The first blocks are

$$P_1 = z, \quad P_2 = \frac{z^2}{2} - z, \quad P_3 = \frac{z^3}{3} - \frac{3}{2}z^2 + z.$$

Thus, with $L_1 = \log z$ and $L_2 = \log L_1$,

$$W(z) = L_1 - L_2 + \frac{L_2}{L_1} + \frac{L_2^2/2 - L_2}{L_1^2} + \frac{L_2^3/3 - \frac{3}{2}L_2^2 + L_2}{L_1^3} + \dots$$

Residual certification

If x_N approximates $f^{-1}(y)$ and $|f'| \geq m > 0$ between x_N and the exact inverse, then

$$|x_N - f^{-1}(y)| \leq \frac{|f(x_N) - y|}{m}.$$

Appendix D

Notation and terminology

Symbol or term	Meaning
X	Large independent variable, normally tending to $+\infty$.
$t = X^{-1}$	Outer small generator.
$L = \log X$	First logarithmic generator.
$\log_2 X$	Iterated logarithm $\log \log X$ when defined.
$\mathbb{K}$	Characteristic-zero coefficient field, commonly $\mathbb{R}$ or $\mathbb{C}$.
$\widehat{\mathcal{PL}}_{\text{poly}}$	Polynomial-coefficient Laurent algebra $\mathbb{K}[L]((t))$.
$\widehat{\mathcal{PL}}$	Iterated Laurent completion, typically $\mathbb{K}((L^{-1}))((t))$.
$\text{ord}_t T$	Least outer exponent with nonzero coefficient.
$\text{supp } T$	Set of monomials or outer exponents occurring in T .
$\prec$	Strict asymptotic dominance: $f/g \rightarrow 0$.
$\circ$	Composition.
$F^{\langle -1 \rangle}$	Compositional inverse of F .
$\mathcal{D}$	Derivative with respect to the large variable.
block	All monomials sharing one outer power t^n .
filtration	Nested ideals or precision levels used to define truncation and coefficientwise convergence.
near identity	A map $F(X) = X + o(X)$ in the selected scale.
series reversal	Compositional inversion, not reciprocal.
summable family	A formal family for which every monomial receives only finitely many contributions.
grid support	Support generated by finitely many small ratios, with well-founded exponent restrictions.
residual	The defining equation evaluated at a truncated approximation.

Concluding perspective

Polynomial–logarithmic transseries occupy a productive middle ground. They retain the ordered-support discipline of general transseries while remaining close enough to ordinary Laurent series for explicit coefficient algorithms. Their apparent complication is concentrated in two facts: logarithms form subordinate scales inside power blocks, and composition moves both the inverse-power generator and the logarithmic generator.

Once those facts are encoded, the main operations become systematic. Multiplication is convolution. Division is triangular. Near-identity composition is either a two-generator substitution or a filtered Taylor series. Reversal is a triangular group operation, accelerated by Newton iteration and summarized in closed form by Lagrange–Bürmann. Lambert W is the canonical example because its logarithmic defining equation is exactly the inverse of $X + \log X$, but the same machinery governs a much wider range of inverse asymptotics.

The most durable computational lesson is to organize every calculation by valuation and to verify every claimed truncation by substitution. The most important conceptual lesson is to separate formal algebra from analytic realization. Formal transseries determine the only possible coefficients; residual estimates, monotonicity, sectorial bounds, or realization theorems then decide when those coefficients describe an actual function.

Bibliography

- [1] R. M. Corless, G. H. Gonnet, D. E. G. Hare, D. J. Jeffrey, and D. E. Knuth, On the Lambert W function, *Advances in Computational Mathematics* **5** (1996), 329–359. <https://doi.org/10.1007/BF02124750>.
- [2] NIST Digital Library of Mathematical Functions, §4.13, Lambert W -function, Version 1.2.7, release date June 15, 2026, accessed August 31, 2026. <https://dlmf.nist.gov/4.13>.
- [3] G. A. Edgar, Transseries for beginners, *Real Analysis Exchange* **35** (2009/2010), no. 2, 253–310. <https://arxiv.org/abs/0801.4877>.
- [4] G. A. Edgar, Transseries: composition, recursion, and convergence, arXiv:0909.1259, 2009. <https://arxiv.org/abs/0909.1259>.
- [5] G. A. Edgar, Fractional iteration of series and transseries, *Transactions of the American Mathematical Society* **365** (2013), 5805–5832. <https://arxiv.org/abs/1002.2378>.
- [6] G. A. Kalugin and D. J. Jeffrey, Convergence in $\mathbb{C}$ of series for the Lambert W function, arXiv:1208.0754, 2012. <https://arxiv.org/abs/1208.0754>.
- [7] P. Mardešić, M. Resman, J.-P. Rolin, and V. Županović, Normal forms and embeddings for power-log transseries, *Advances in Mathematics* **303** (2016), 888–953. <https://doi.org/10.1016/j.aim.2016.08.030>.
- [8] B. Salvy and J. Shackell, Symbolic asymptotics: multiserries of inverse functions, *Journal of Symbolic Computation* **27** (1999), no. 6, 543–563. <https://doi.org/10.1006/jsco.1999.0281>.
- [9] J. Shackell, *Symbolic Asymptotics*, Springer, Berlin, 2004.
- [10] L. van den Dries, A. Macintyre, and D. Marker, Logarithmic-exponential series, *Annals of Pure and Applied Logic* **111** (2001), 61–113. [https://doi.org/10.1016/S0168-0072\(01\)00035-5](https://doi.org/10.1016/S0168-0072(01)00035-5).
- [11] J. van der Hoeven, *Transseries and Real Differential Algebra*, Lecture Notes in Mathematics, vol. 1888, Springer, Berlin, 2006. <https://doi.org/10.1007/3-540-35590-1>.